The Grounding Paradox: When True Physical Inputs Make a Model Worse, and How to Locate a Physics Prior's Ceiling

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Abstract

Physics-informed models increasingly route a prediction through a fixed, differentiable physical *closure*—a mechanistic map with no fitted parameters—fed by a learned intermediate. For solid–liquid solubility the closure is COSMO-SAC, which turns a molecule's surface-charge σ-profile into an activity coefficient; the σ-profile is predicted by a network. The premise is that grounding the closure on truer physical inputs improves it. We show it can do the opposite: replacing the model's learned σ-profiles with the true, externally measured ones (VT-2005) makes activity and solubility prediction *worse* than a free learned head. We turn this anomaly into a general method for locating a physics prior's ceiling. Because the physical input has external ground truth, the true-input error splits—with no model fit—into a part the inputs cannot resolve and a part the fixed closure gets wrong; on our activity data the closure is the binding term. We give the supporting theory for any composed predictor: the crystal/activity split is non-identifiable (a free head silently compensates), a grounding-gap identity, and a sign rule for the "physics tax" that predicts a critical data budget beyond which such a prior must lose. We are careful about reach—the grounding gap only corroborates the decomposition, which does the certifying—and the decomposition transfers to synthetic closure families and a second chemical closure (pKa through a Hammett relationship). Across every training budget we test, the physics prior gives no accuracy gain. What it buys is an interpretable decomposition and calibrated uncertainty, not lower error.

1 Introduction

Coupling a graph encoder to an explicit thermodynamic bottleneck is a common recipe for out-of-distribution solubility prediction: the model must route its prediction through interpretable physical factors, and each factor can in principle be grounded on abundant single-component data that a black-box predictor cannot exploit. The expectation is that the closer the model's physical inputs are to the truth, the better it should predict. On the activity branch this expectation fails. We characterize the failure quantitatively, and it is the characterization, more than the failure itself, that this paper is about. That true σ-profiles can hurt a COSMO-SAC pipeline will not surprise a thermodynamics specialist who knows the model's limits; the contribution is not the failure but an externally grounded procedure that *measures* where such a physics prior's ceiling lies and attributes it to the closure rather than the inputs. The procedure is general—we state it for any composed predictor $\hat{y} = g(h(x))$ with a fixed closure over a learned intermediate, prove its core, and reproduce it on synthetic closure families (§6.3) and a second chemical closure (pKa through a Hammett relationship, §6.5). What stays regime-specific is the *magnitude* of closure-dominance we measure for COSMO-SAC on low-γ activity data; whether that particular ordering persists across solubility chemistries we scope honestly (§7).

Contributions.

- **The grounding paradox (empirical).** Substituting true VT-2005 σ-profiles into the differentiable COSMO-SAC closure degrades activity ($\ln \gamma_2^\infty$) and solubility ($\ln x_2$) prediction relative to a free learned activity head, the opposite of the intended benefit (§6.1).
- **A measurement affordance: the closure/insufficiency split.** Because the physical latent $z^\star$ (the true σ-profile) has external ground truth, the true-input excess risk splits *exactly* as $B = B_{\text{closure}} + B_{\text{insuff}}$ with no model fit (Lemma 3). The decomposition and its estimators are textbook; the new element is exploiting an externally measured latent to bound the split empirically. On the convention-robust evidence—a bound that holds however the COSMO-SAC closure is configured, plus the more-physics-worse effect—B_{closure} (misspecified decoder) exceeds B_{insuff} (input insufficiency) on the low-γ subset we can measure, robustly to the insufficiency estimator and to leaving out any solvent: there, the binding term is the closure (§6.2).
- **A negative actionability result (Gate B).** A post-hoc output-space residual placed on the physics prediction recovers calibration but not accuracy; the closure binding cannot be undone downstream (§6.4).
- **A broad, explained negative.** The physics prior does not beat a matched black box on accuracy at any training-data budget—including the low-data regime where an inductive bias should help most (§6.8)—nor, as a trained model, on temperature (though the physical hypothesis *class* does extrapolate there, §6.10); read against the lemmas and the closure measurement, this points to a *structural* ceiling within this closure family rather than an incidental one (§7). We make no accuracy claim.
- **A general account, and honest scoping.** The method is not tied to COSMO-SAC: it is stated for any composed predictor $\hat{y} = g(h(x))$ with a fixed closure g over a learned intermediate. We prove three structural lemmas (Lemmas 1–3), a grounding-gap identity (Theorem 1), and a sign rule for the physics tax (Prop. 1). The grounding-gap identity bounds the gap by the closure bias, but only under assumptions that fail here; we therefore read the gap as a *symptom* and let the assumption-light decomposition do the certifying. The sign rule's critical-budget and aleatoric-floor corollaries predict the two negatives above. The *decomposition* transfers to synthetic closure families (§6.3) and a second chemical closure—pKa through a Hammett relationship (§6.5; the trained-model paradox and tax there are left to future work)—and all proofs are elementary (App. B).

2 Related work

Two families of methods now dominate solid solubility and solvation prediction, and they make opposite bets about physical structure. Physics-informed pipelines such as SolProp [25] assemble a thermodynamic cycle from separately learned components (solvation free energy, vapor pressure, fusion properties), so that each prediction is decomposable and the intermediate quantities carry physical meaning. Direct models forgo that structure: FastSolv [1] regresses solubility from molecular descriptors with a descriptor-based deep neural network (fastprop), and it is the current leader on new-solute extrapolation, a task on which inter-laboratory disagreement sets a substantial irreducible-error floor. The ordering is unambiguous on accuracy: the black box leads, and the physics-informed cycle trails it. Our work does not try to overturn this ordering. It asks instead where the ceiling of the physics prior comes from, and it uses a matched direct control that shares our encoder so that the accuracy gap is attributable to the closure rather than to representation.

The activity coefficient is the term our closure must supply, and learned surrogates for it have advanced quickly. Winter et al. [29] predict infinite-dilution activity coefficients $\ln \gamma_2^\infty$ directly from SMILES with a language model, and Sanchez-Medina et al. [22] embed a Gibbs-Helmholtz constraint in a graph network to capture the temperature dependence of $\ln \gamma_2^\infty$. These treat $\ln \gamma_2$ as the regression target. We take the complementary route of retaining a mechanistic activity model, either COSMO-SAC over a predicted σ-profile or NRTL, as a differentiable closure inside the solver, so that the network predicts physical inputs rather than the coefficient itself. That choice is what exposes the failure mode we measure: the misspecification of a fixed mechanistic map can dominate the insufficiency of its learned inputs.

Beyond solubility, the question we pose—when a fixed physical structure helps or hurts a learned model—is the central tension of physics-informed machine learning [9]. Its dominant instantiations impose structure as a differentiable object: physics-informed neural networks enforce a governing equation through the loss [18], while physics-guided and gray-box hybrids compose learned components with fixed mechanistic maps [28]—exactly our $g(h(x))$, with g the mechanistic map. The prevailing expectation is that more physics improves generalization, yet the same structure can degrade it: physics-informed networks have documented failure modes in which the imposed constraint makes optimization or accuracy worse [12]. Our contribution is to make that degradation *measurable*—to decompose it into closure misspecification and input insufficiency and to give a sign rule for when a fixed-closure prior must lose—rather than to catalogue it case by case; the solubility closure is our worked instance, and the pKa closure and synthetic families show the account is not tied to it.

We borrow several diagnostic lenses and claim no new theory from them. The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O'Hagan [10] and its calibration critique [2]; separating whether predictions are right on average from whether they are sharp is the calibration refinement and reliability-resolution decomposition of Murphy and DeGroot [17, 4]; fitting a wrong model still converges to a well-defined pseudo-true limit under the quasi-maximum-likelihood account of White [27]. That the decomposition is only weakly constrained by solubility alone is a matter of parameter sloppiness [23, 7] and of structural versus practical identifiability [20, 26]. Finally, the risk that apparent skill reflects pipeline choices rather than physics is the concern of underspecification [3], leakage and reproducibility [8], and compositional risk [14]. Our data come from BigSolDB 2.0 [11].

Positioned against this literature, our contribution is not a new model and not an accuracy result. It is a method—with a proved core (three structural lemmas, a grounding-gap theorem, and a sign rule for the "physics tax") and a reusable measurement—for locating where a physics prior's accuracy ceiling lives, stated for any composed predictor rather than one model: we decompose the true-input error into a closure-misspecification part B_{closure} and an input-insufficiency part B_{insuff}, and show that feeding true physical inputs through the closure can make predictions worse when B_{closure} dominates. What the physics buys is not accuracy but an interpretable decomposition, an account of its own limiting term, and calibrated uncertainty, while FastSolv and SolProp remain the reference points for accuracy.

3 Problem setup and notation

For a solute s in solvent v at temperature T, the target is $y = \ln x_2$, the log saturated mole fraction. Dissolving a solid trades two costs against each other—breaking the crystal lattice (Φ)

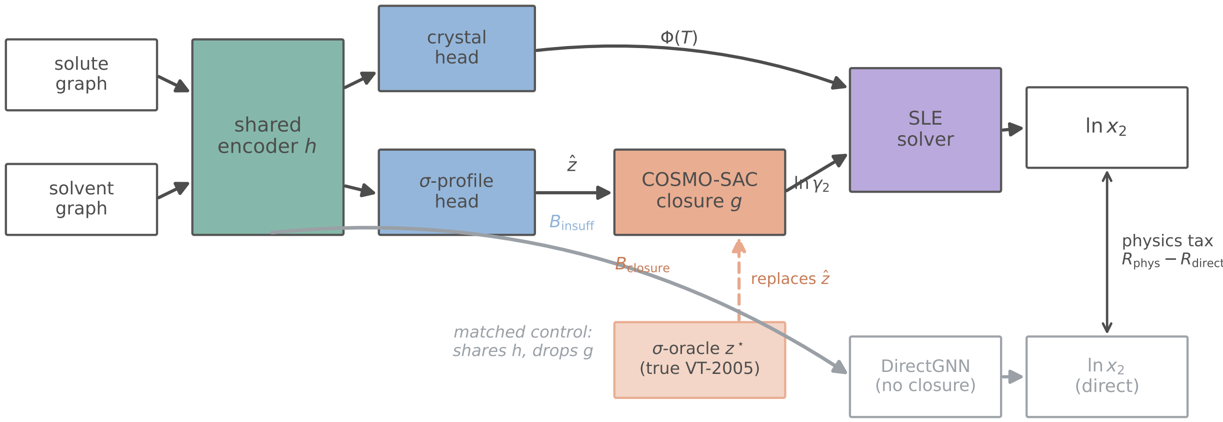


Figure 1: The model and the grounding intervention. Solute and solvent graphs pass through a shared encoder to a crystal head (the ideal term $\Phi(T)$) and a σ-profile head; the predicted profile $\hat{z}$ feeds the fixed COSMO-SAC closure g, which returns $\ln\gamma_2$, and an SLE solver combines the two into $\ln x_2$. Input insufficiency B_{insuff} enters at the intermediate $\hat{z}$ and closure misspecification B_{closure} at g. The σ-oracle replaces $\hat{z}$ with the true VT-2005 profile $z^\star$ (salmon). DirectGNN (grey) is the matched control—it shares the encoder but drops the closure, so the gap to the physics path is the physics tax $R_{\mathrm{phys}} - R_{\mathrm{direct}}$.

and mixing the freed molecules against non-ideal solute–solvent interactions ($\ln\gamma_2$)—and solid–liquid equilibrium (SLE) sets the log-solubility to their negative sum:

$$\ln x_2 = -\underbrace{\frac{\Delta H_{\mathrm{fus}}}{R}\left(\frac{1}{T} - \frac{1}{T_m}\right)}_{\Phi(T)\ \text{(crystal, depends on } s)} - \underbrace{\ln\gamma_2(x_2, T)}_{\text{activity, depends on }(s,v)}, \tag{1}$$

with the constant-ΔC_p (heat-capacity-of-fusion) correction omitted for clarity. The activity term is produced by a *differentiable closure* g: a graph encoder predicts a σ-profile—the distribution of screening (surface) charge density over a molecule, the histogram a COSMO model consumes—for each molecule, and COSMO-SAC (the COnductor-like Screening Model–Segment Activity Coefficient [13]) maps the pair of profiles to $\ln\gamma_2$. Write $z^\star$ for the *true* physical latent, the externally measured σ-profile from the VT-2005 database (the Virginia Tech 2005 compilation of quantum-chemically computed screening-charge profiles for ~1,400 compounds [16]), and $g(z^\star)$ for the closure output when fed the true profile. The model's own head instead predicts a *learned* profile $\hat{z}$ and evaluates $g(\hat{z})$; a matched *DirectGNN* control shares the encoder but emits $\ln x_2$ with no closure, and is the reference for "what the physics bottleneck costs" (Fig. 1).

The grounding intervention. Feeding $z^\star$ in place of $\hat{z}$ (a σ-oracle) isolates the closure from encoder error: if the closure were well specified, $g(z^\star)$ should be the best available activity estimate. The paradox is that it is not.

Table 1: Notation and coined terms used throughout.

Symbol / term	Meaning
g (closure)	fixed, non-fitted physical map (COSMO-SAC; a Hammett LFER for pKa)
h (encoder)	learned graph network, $x \mapsto z$
$z^\star$, $\hat{z}$	true (oracle) vs. learned physical intermediate (the σ-profile)
m	activity target (experimental $\ln \gamma_2^\infty$)
B_{closure}	closure misspecification, $\mathbb{E}[(\mathbb{E}[m \mid z^\star] - g(z^\star))^2]$
B_{insuff}	input insufficiency, $\mathbb{E}[\text{Var}(m \mid z^\star)]$
Γ (grounding gap)	$R_{\text{orc}} - R_{\text{e2e}}$: oracle risk minus best end-to-end risk
Δ_∞ (physics tax)	asymptotic risk of the physics arm minus the matched control
full / res	COSMO-SAC with / without the Staverman–Guggenheim combinatorial term
Tier 1 / Tier 2	convention-independent (load-bearing) / convention-robust corroboration

Data. Solubility measurements are a Bemis–Murcko scaffold split of BigSolDB [11] (grouping molecules by ring-system-plus-linker scaffold, side chains stripped), so test solutes share no scaffold with training; the activity labels m are experimental infinite-dilution activity coefficients ($\ln \gamma_2^\infty$, IDAC) compiled from ThermoML (the NIST/IUPAC archive of experimental thermophysical data [6]), and the true σ-profiles $z^\star$ are the VT-2005 database.

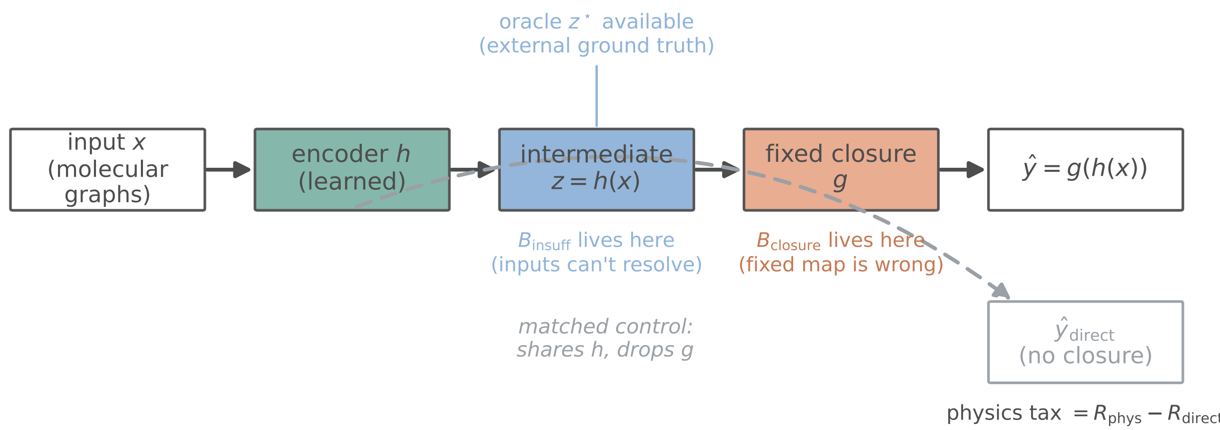
4 Theory: locating a physics prior's ceiling in a composed predictor

The results below are stated for a general *composed predictor* $\hat{y} = g(h(x))$ in which a fixed, non-fitted map g—the *closure*—consumes a learned physical intermediate $z = h(x)$, and are compared against a matched free predictor that shares the encoder but drops g. Solubility, with g the COSMO-SAC closure and z the σ-profile, is one instance; §6.5 gives a second (pKa through a fixed Hammett closure), and §6.3 a controlled synthetic family. Nothing in Lemmas 1–3, Theorem 1, or Proposition 1 is specific to COSMO-SAC. Two quantities organize the section: the closure/insufficiency split $B = B_{\text{closure}} + B_{\text{insuff}}$, which is the assumption-light *instrument*, and the observable grounding gap Γ, which is a *symptom* of closure misspecification whose validity we characterize (Fig. 2).

4.1 Why a free head can compensate: structural non-identifiability

The crystal/activity split is not identifiable from solubility data alone, which is precisely why an unconstrained head can silently absorb crystal error into the activity term (Fig. 3).

Lemma 1 (Structural non-identifiability of the SLE split)**.** *Consider the infinite-dilution SLE observable $\mu(T) = -\Phi(T) - \ln \gamma_2^\infty(T)$ with $\Delta C_p = 0$, so that $\Phi(T) = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$ is affine in $u = 1/T$, and a symmetric two-suffix activity class ($\ln \gamma_2 = \ln \gamma_2^\infty (1 - x_2)^2$) in which $\ln \gamma_2^\infty(T)$ is affine in u. Then, for any design with at least two distinct temperatures, μ depends on the parameters only through the two-dimensional statistic (its slope and intercept in u), and the activity intercept and slope already span that plane; hence the crystal parameters $(\Delta H_{\text{fus}}, T_m)$ are unidentified, the profiled Fisher information for ΔH_{fus} is exactly zero, and the indeterminacy set is two-dimensional. The deficiency is closed only by a rank-completing external constraint (a direct crystal label, or a fixed activity slope). At finite dilution the composition factor $(1 - x_2)^2$ breaks*



the same abstraction, three instances

	input x	**encoder** h	**intermediate** z	**fixed closure** g	**target** y
solubility	solute+solvent graphs	MPNN	σ-profile	COSMO-SAC	$\ln x_2$
pKa	molecular graph	MPNN	Hammett σ	LFER $pK_{a0} - \rho\sigma$	pK_a
synthetic	features	MLP	latent z	deformed teacher	m

Figure 2: The general composed predictor $\hat{y} = g(h(x))$ and its matched control. A learned encoder h produces a physical intermediate z; a *fixed* closure g maps it to the prediction. Input insufficiency B_{insuff} lives in z (what the intermediate cannot resolve) and closure misspecification B_{closure} in g (where the fixed map is wrong); an external oracle $z^\star$ lets us measure the split. The matched control (grey) shares h but drops g, so the gap between them is the "physics tax." The same abstraction instantiates as solubility, pKa, and the synthetic dial (table).

the exact affinity, leaving a non-zero but near-degenerate information of order x_2^2 *(numerically, condition number* 2.3×10^5*; §6.6).*

This is the standard structural-identifiability picture for a collinear design [20, 26]; we prove it in the SLE form (App. B) and audit it numerically in §6.6. It is *enabling background*, not our headline: it explains why grounding an under-determined factor on external single-component data is necessary, and why compensation between the factors is expected. By itself it says nothing about whether grounding helps. That depends on whether the closure that consumes the grounded inputs is correctly specified, which is the question we make measurable next.

The non-identifiability is not binary but graded by how much of the crystal direction the activity class can absorb.

Lemma 2 (Class-dependent efficient information)**.** *Let the activity nuisance lie in a class* $\mathcal{H}$ *with tangent projection* $\Pi_{\mathcal{H}}$*, and let* $v = \partial\mu/\partial\Delta H_{\text{fus}}$ *be the crystal score. The efficient information for* ΔH_{fus} *is* $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = \sigma^{-2}\,\|v - \Pi_{\mathcal{H}} v\|^2$*. If* $\mathcal{H}$ *contains the crystal score's span* $\{1, 1/T\}$ *(a free intercept and slope) then* $\Pi_{\mathcal{H}} v = v$ *and* $\mathcal{I}_{\text{eff}} = 0$*, recovering Lemma 1; if* $\mathcal{H}$ *is restricted (the slope is known or constrained) then* $\mathcal{I}_{\text{eff}} > 0$ *and the Cramér–Rao lower bound (CRLB) on* ΔH_{fus} *is finite.*

In words: the efficient information is what the data can still pin down about the crystal term *after* the activity model has absorbed everything within its reach. If that model can fit a free

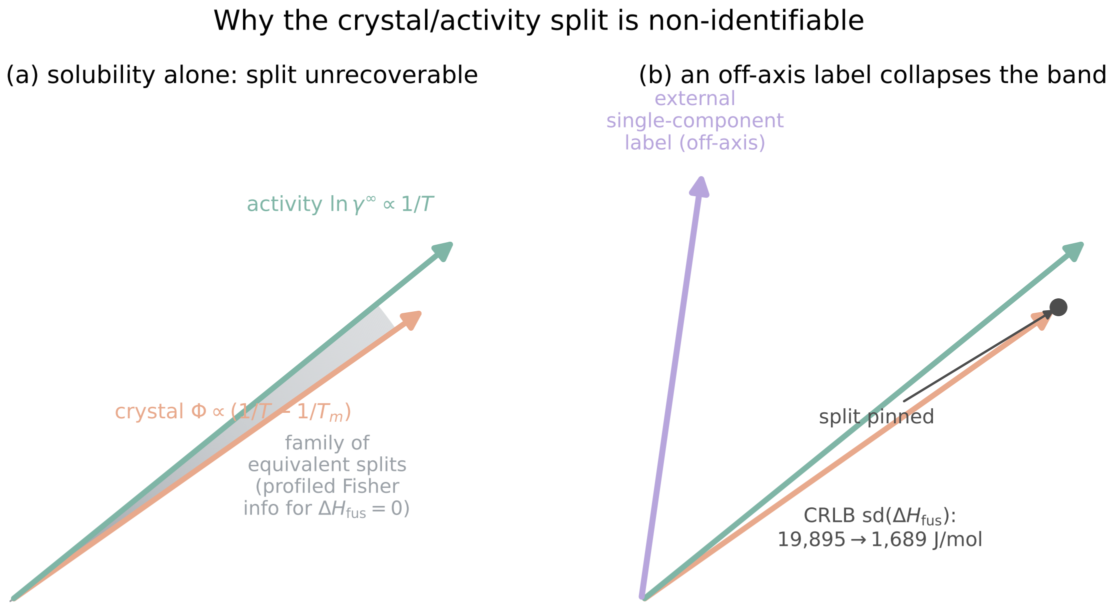


Figure 3: Why the crystal/activity split is non-identifiable from solubility. *(a)* As functions of $1/T$ the crystal direction Φ and the activity direction $\ln\gamma^\infty$ are nearly collinear, so a whole band of crystal/activity splits fits the data equally well (the profiled Fisher information for ΔH_{fus} is zero, Lemma 1). *(b)* An external single-component label points off that axis and collapses the band to a point, pinning the split—the mechanism behind the aux-stream grounding, with the measured Cramér–Rao drop on ΔH_{fus}.

intercept and slope in $1/T$, it absorbs the entire crystal direction and nothing identifies ΔH_{fus} ($\mathcal{I}_{\text{eff}} = 0$); constraining the slope leaves a little information behind. The proof is the efficient-score construction (App. B): project the crystal score onto the orthogonal complement of the nuisance tangent space; the squared norm of the residual is the efficient information. The numerical audit of §6.6 quantifies both regimes, and shows the honest corollary: merely *restricting* the activity slope leaves $\text{sd}(\Delta H_{\text{fus}}) \approx 8{,}251$ J/mol — finite but unusable — whereas an *external* crystal label, which contributes a score direction outside the activity tangent space, closes it to $\approx 1{,}689$ J/mol.

4.2 The closure–insufficiency decomposition

Let m be the activity target (experimental $\ln\gamma_2^\infty$) and $g(z^\star)$ the closure evaluated on the true latent. The excess risk of the true-input path over the best-possible predictor $\mathbb{E}[m \mid \cdot]$ (the Bayes predictor) decomposes exactly—into a part the intermediate cannot resolve and a part the fixed map gets wrong (Fig. 4).

Lemma 3 (Exact closure–insufficiency decomposition and its one-sided bounds)**.** *For a* fixed *(non-fitted) closure g,*

$$\underbrace{\mathbb{E}\big[(m - g(z^\star))^2\big]}_{\textit{MSE of the true-input path}} = \underbrace{\mathbb{E}\big[\mathrm{Var}(m \mid z^\star)\big]}_{B_{\text{insuff}}\ \textit{(inputs cannot resolve)}} + \underbrace{\mathbb{E}\big[(\mathbb{E}[m \mid z^\star] - g(z^\star))^2\big]}_{B_{\text{closure}}\ \textit{(decoder mis-map)}}. \tag{2}$$

The cross term vanishes identically because $\mathbb{E}[m \mid z^\star] - g(z^\star)$ is $z^\star$-measurable and $\mathbb{E}[(m - \mathbb{E}[m \mid z^\star]) \mid z^\star] = 0$. Both components are separately estimable from data with no model fit*, because $z^\star$ is externally observed.*

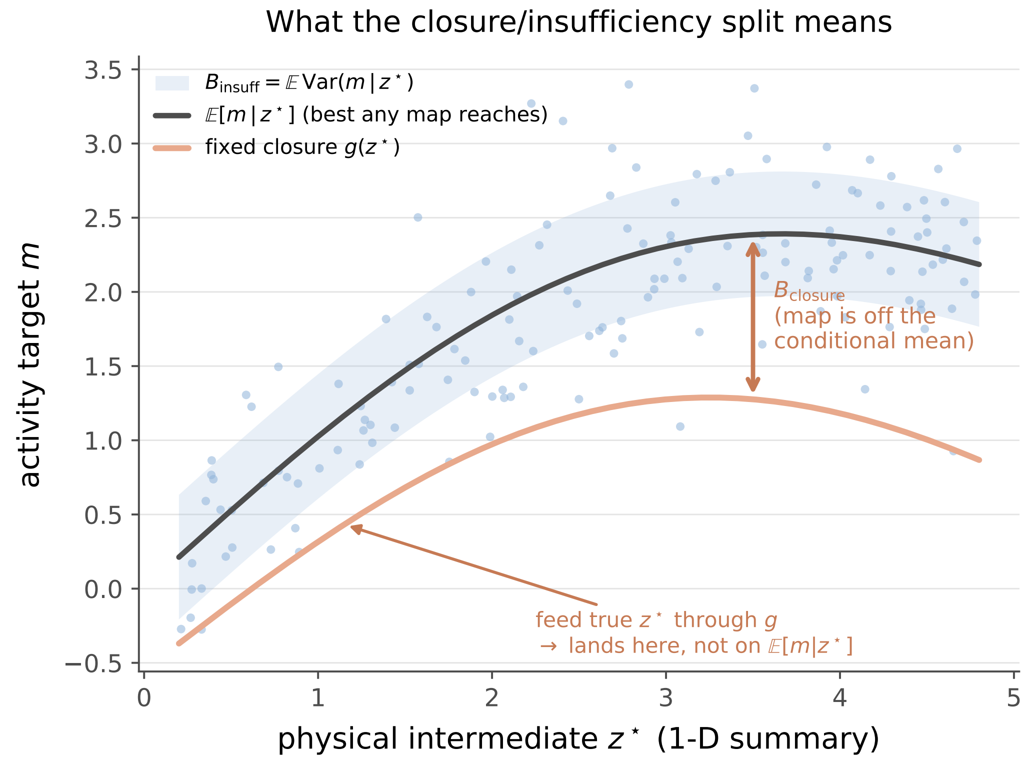


Figure 4: What the closure/insufficiency split means. The activity target m is scattered against a one-dimensional summary of the physical intermediate $z^\star$. The vertical spread of the conditional cloud is the input insufficiency $B_{\text{insuff}} = \mathbb{E}\,\text{Var}(m \mid z^\star)$—what no map of $z^\star$ can resolve—while the gap between the conditional mean $\mathbb{E}[m \mid z^\star]$ (the best any map reaches) and the fixed closure $g(z^\star)$ is the closure misspecification B_{closure}. Feeding the true $z^\star$ through g lands on the lower curve, off the conditional mean; that displacement is the paradox. Schematic; the measured bounds are Fig. 7.

Equation (2) is a bias–variance split of model discrepancy in the sense of Kennedy–O'Hagan [10] and Brynjarsdóttir–O'Hagan [2]: B_{closure} is the systematic discrepancy of a structurally fixed physical map, and its dominance over B_{insuff} is the empirical content of "the ceiling is the closure, not the inputs." We do not claim (2) as new; the affordance is that the external groundedness of $z^\star$ lets us *measure* the split rather than assume it. **The exactness is contingent on g being fixed**: for a *fitted* decoder the cross term does not vanish and a plug-in point decomposition is invalid (we observe a spurious -0.36 cross term when $\mathbb{E}[m \mid z^\star]$ is replaced by a fitted estimate). We therefore report *one-sided bounds*, not a point split (each proved in App. B):

- $B_{\text{closure}} \geq (\mathbb{E}[m] - \mathbb{E}[g])^2$ by Jensen: a constant offset between the decoder's mean output and the target is misspecification, not input insufficiency, and this bound needs *no smoothness assumption*. Its one requirement is that m and g share the $\ln \gamma_2^\infty$ reference state, which we check directly (§6.2).
- $B_{\text{insuff}} \leq \mathbb{E}[\text{Var}(m \mid \text{bin}(g(z^\star)))]$, a valid *upper* bound (law of total variance, since a binning of $g(z^\star)$ coarsens $z^\star$).

Because $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^\star)]$ depends only on $(m, z^\star)$, it is *independent of the closure convention*: any disagreement between closure variants is decoder disagreement and lands in B_{closure}.

4.3 The grounding gap: a symptom of closure misspecification

The paradox—true inputs hurting—is a measurable quantity, not a curiosity. Let $\mathcal{E}$ be the encoder class ($h : x \mapsto z$) and define three population risks of the activity target m: the oracle risk $R_{\mathrm{orc}} = \mathbb{E}[(m - g(z^\star))^2]$, the best end-to-end risk $R_{\mathrm{e2e}} = \inf_{h\in\mathcal{E}} \mathbb{E}[(m - g(h(x)))^2]$, and the free-predictor (black-box) risk $R_{\mathrm{free}} = \mathbb{E}[(m - \mathbb{E}[m \mid x])^2]$. The *grounding gap* is $\Gamma := R_{\mathrm{orc}} - R_{\mathrm{e2e}}$; the paradox is $\Gamma > 0$ (a free learned head beats the true input pushed through the closure).

Theorem 1 (Grounding-gap sandwich)**.** *Assume all risks are finite ($m, g(z^\star) \in L^2$; g and each $h \in \mathcal{E}$ measurable), (A1) the true encoder is representable, $\phi^\star \in \mathcal{E}$ with $z^\star = \phi^\star(x)$, and (A2) the intermediate is a lossless bottleneck, $\mathbb{E}[m \mid x] = \mathbb{E}[m \mid z^\star]$. Then*

$$0 \;\le\; \Gamma \;\le\; B_{\mathrm{closure}}, \qquad \Gamma = B_{\mathrm{closure}} - \big(R_{\mathrm{e2e}} - R_{\mathrm{free}}\big), \tag{3}$$

with $\Gamma = B_{\mathrm{closure}}$ iff $R_{\mathrm{e2e}} = R_{\mathrm{free}}$ (the composed class realizes the Bayes predictor by distorting the intermediate). In particular a well-specified closure ($B_{\mathrm{closure}} = 0$) gives $\Gamma = 0$; hence under A1–A2*, $\Gamma > 0$ certifies $B_{\mathrm{closure}} > 0$.*

The proof is elementary (App. B): $\Gamma \ge 0$ because $\phi^\star$ is a feasible encoder; $R_{\mathrm{e2e}} \ge R_{\mathrm{free}}$ because every $g(h(x))$ is $\sigma(x)$-measurable; and $R_{\mathrm{free}} = B_{\mathrm{insuff}}$ under A2. Thus Γ is exactly the closure bias the encoder can *absorb* by feeding g a wrong-but-useful intermediate—the formal content of Lemma 1's "a free head silently compensates."

Γ is a symptom, not a certificate. Both assumptions are load-bearing and neither holds cleanly here. A1 is *empirically false*: the encoder is transfer-limited (linear-probe test $R^2 \approx 0.45$), so $\phi^\star \notin \mathcal{E}$ and even $\Gamma \ge 0$ is not guaranteed. A2 is unverifiable without the oracle and physically dubious (the σ-profile is a lossy summary of the graph). When A2 fails, $\Gamma > 0$ *conflates* two causes: closure misspecification ($B_{\mathrm{closure}} > 0$) and input insufficiency (a correct closure driven off-support to recover information the intermediate discarded). A controlled construction with a *well-specified* closure and a lossy intermediate gives $B_{\mathrm{closure}} = 0$ yet $\Gamma > 0$ (§6.3, and the pKa ortho sweep of §6.5). Certification of closure-dominance therefore rests on the convention-independent *decomposition* (Lemma 3), which needs neither A1 nor A2: in solubility the Jensen bound and the insufficiency estimates place B_{closure} above B_{insuff} *directly* (§6.2), and the grounding gap only corroborates it. This is the paper's method: *the decomposition is the instrument, Γ is the cheap symptom that motivates it, and separating its two causes is what an externally measured latent buys.*

4.4 The physics tax and the critical data budget

The gap is a population statement; the deployed question is whether the *trained* physics model beats the *trained* black box at sample size n. Writing each expected risk as an asymptote plus estimation variance, $\mathbb{E}[\hat{R}_{\mathrm{phys}}(n)] \approx R_{\mathrm{e2e}} + V_{\mathrm{phys}}(n)$ and $\mathbb{E}[\hat{R}_{\mathrm{direct}}(n)] \approx R_{\mathrm{direct},\infty} + V_{\mathrm{direct}}(n)$ with $V(n) \downarrow 0$, the *physics tax* is

$$T(n) \approx \Delta_\infty + \big[V_{\mathrm{phys}}(n) - V_{\mathrm{direct}}(n)\big], \qquad \Delta_\infty := R_{\mathrm{e2e}} - R_{\mathrm{direct},\infty}. \tag{4}$$

Proposition 1 (Sign of the physics tax)**.** *(i) $T(n) < 0$ (physics helps) iff $V_{\mathrm{direct}}(n) - V_{\mathrm{phys}}(n) > \Delta_\infty$: the variance a constrained prior saves must exceed its asymptotic bias. (ii) If $\Delta_\infty > 0$ and the variance gap vanishes ($V_{\mathrm{direct}} - V_{\mathrm{phys}} \to 0$), there is a critical budget $n^\star$ beyond which $T(n) > 0$ for all $n > n^\star$. (iii) Since $V_{\mathrm{direct}} - V_{\mathrm{phys}} \le V_{\mathrm{direct}}$, as the black box nears the aleatoric floor ($V_{\mathrm{direct}} \to 0$) the tax tends to $\Delta_\infty \ge 0$.*

When the control is asymptotically Bayes and A2 holds, $\Delta_\infty = B_{\text{closure}} - \Gamma$, tying the asymptotic tax to the *un-compensatable* part of the closure bias; in general Δ_∞ is read off the large-n asymptote. Both corollaries are borne out: the data-efficiency null—a positive tax at *every* budget, growing with n (§6.8)—is case (ii) with Δ_∞ above the largest variance gap, and the novelty inversion—the tax heaviest on the *best-covered* solutes (§6.9)—is case (iii). We prove the structural parts (existence of $n^\star$ given $\Delta_\infty > 0$ and a vanishing variance gap; the floor inequality); the shapes of $V(n)$ are empirical.

4.5 Supporting structural facts

Three further consequences of the same collinear geometry are used informally and are not stated as theorems. *(i) Temperature extrapolation.* Within the affine-in-$1/T$ activity family, same-pair prediction at a temperature $T^\star$ outside the training range is an affine extrapolation whose error is controlled by the fitted-slope variance times $|1/T^\star - \overline{1/T}|$, whereas an unstructured temperature encoder is unconstrained beyond its support. This is the one place the physical *structure* is expected to help; §6.10 reports the class-level margin, which is a property of the hypothesis class and not an accuracy claim for the trained model (indeed the trained models trail the van't Hoff baseline). *(ii) Geometry of compensation.* As the training window $\Delta T \to 0$, $1/T$ becomes nearly constant, the crystal and activity design columns become collinear, and the smallest Fisher eigenvalue tends to zero (the ΔT sweep of §6.6 reaches conditioning 10^{16} at $\Delta T = 10\,\mathrm{K}$), so an optimizer prefers to split error between the factors rather than fit either correctly. *(iii) Grounding design.* Each single-component stream rank-completes a different deficient direction of Lemma 1; this is the design rule behind the auxiliary streams, though (as §6.7 shows for a dose-limited crystal run) it does not by itself promise an end-task gain.

4.6 What we prove, and what we do not claim

We prove the three structural lemmas the measurement rests on — the split's non-identifiability (Lemma 1), its class-dependence (Lemma 2), and the exact closure/insufficiency decomposition with its one-sided bounds (Lemma 3) — together with the grounding-gap sandwich (Theorem 1) and the sign rule for the physics tax (Prop. 1); all proofs are elementary (App. B). We are deliberate about their reach. Theorem 1 certifies closure misspecification *only* under A1–A2, and both fail or are unverifiable in our setting (A1 empirically, A2 in principle); we therefore rest closure-dominance on the assumption-light decomposition and use Γ as a corroborating symptom, not a certificate. Proposition 1 predicts the *sign* of the tax from measurable quantities, not an optimum. In particular there is no "optimal grounding" theorem and no necessary-and-sufficient condition for the decomposition to improve out-of-distribution accuracy — the honest empirical answer (§6) is that it does not. The lemmas are read against, not in competition with, known results on model discrepancy [10, 2], parameter sloppiness [23, 7], structural and practical identifiability [20, 26], estimation under misspecification [27], and underspecification [3]; the calibration-versus-accuracy separation of §6.4 is the classical refinement decomposition of a probability forecast [17, 4].

5 Methods

The model is a controlled comparison: a shared graph backbone feeds either an explicit thermodynamic bottleneck (*TGNNSolv*) or a direct regressor (*DirectGNN*) that shares the backbone but emits $\ln x_2$ with no physics. The gap between them measures what the bottleneck costs or buys.

This section describes the shared backbone, the three interchangeable activity closures, the solid–liquid-equilibrium (SLE) solver and its implicit differentiation, the grounding intervention, and the training protocol. Exact hyperparameters, the full feature list, and the complete loss taxonomy are in the SI (§A).

5.1 Data, splits, and labels

Solubility measurements are drawn from BigSolDB 2.0 [11] and partitioned by Bemis–Murcko scaffold (`scripts/prepare_data.py`) so that every test solute is unseen at training; row membership is set by a melting-point-independent miscibility/structure filter, and crystal T_m, ΔH_{fus}, Hansen parameters, and infinite-dilution activity coefficients $\ln \gamma_2^{\infty}$ are merged on as per-solute labels where available. Two data hazards are handled explicitly. Melting-point tables arrive in either °C or Kelvin and are auto-detected by column median (a past double-conversion inflated every T_m by 273 K, which propagates directly into the ideal term); and the seeded scaffold split is not stable across pipeline versions, so every regeneration is treated as a new split with all downstream artifacts recomputed. Crystal supervision is scarce: the test and validation splits contain essentially no measured ΔH_{fus}, which is the quantitative backbone of the identifiability argument (§4). External single-component data used for grounding are an open crystal-property pool (~15,000 melting points), the VT-2005 σ-profile database, and ThermoML infinite-dilution activity coefficients; each is filtered to exclude any molecule whose scaffold appears in the validation or test split.

5.2 Graph encoder and pair representation

Solute and solvent are encoded by a *shared* graph network with three interchangeable backbones: a message-passing network (MPNN), a hybrid local-message/global-attention graph transformer (GPS [19]), and a physics-channelled variant, Thermodynamically-Informed Message Passing (TIMP), that splits messages into dispersive, polar, and hydrogen-bonding channels supervised toward the Hansen solubility parameters (the dispersive, polar, and hydrogen-bonding components of a molecule's cohesive energy density) to prevent channel collapse.[1] Small hydride solvents such as water are given explicit hydrogens so the polar and hydrogen-bond channels activate. Solute and solvent embeddings interact through a bipartite cross-attention/message-passing block and a physics-aware readout, and are combined into a pair representation $[\, g_s,\, g_v,\, g_s \odot g_v,\, |g_s - g_v|\,]$ passed through an MLP, optionally augmented with RDKit descriptors, Morgan fingerprints, and an ionic-context feature. All heads below consume this pair representation; the DirectGNN control consumes the identical representation.

5.3 Crystal head and the ideal term

A fusion head predicts the two crystal properties that define the ideal (activity-free) solubility, with physically bounded parametrisations $T_m = 100 + 600\,\sigma(\cdot) \in [100, 700]\,\text{K}$ and $\Delta H_{\text{fus}} = S_H\,\text{softplus}(\cdot) > 0$; an optional entropy-coupled mode instead predicts a Walden-constrained fusion entropy (Walden's rule: rigid organics have a near-constant entropy of fusion, $\sim 57\,\text{J}\,\text{mol}^{-1}\,\text{K}^{-1}$) and sets $T_m = \Delta H_{\text{fus}}/\Delta S$. When a group-contribution T_m prior is used it is affine-calibrated on the training set before entering the head. The crystal properties give the ideal term

$$\Phi(T) = \frac{\Delta H_{\text{fus}}}{R}\left(\frac{1}{T} - \frac{1}{T_m}\right), \tag{5}$$

[1] TIMP is a physics-motivated backbone we introduced to test whether a thermodynamically structured encoder helps; across our runs it does not beat the plain MPNN, consistent with the encoder being transfer-limited rather than expressivity-limited (a linear probe reaches train $R^2 \approx 0.76$ but test $R^2 \approx 0.45$).

the constant-ΔC_p correction omitted (its magnitude is audited in the SI, §E).

5.4 Three interchangeable activity closures

The activity term $\ln \gamma_2$ is produced by one of three closures, selected by configuration, all differentiable and all consuming the same pair representation.

NRTL (Non-Random Two-Liquid [21]). A head predicts the two interaction energies and a non-randomness factor, $\tau_{12}^{\text{ref}}, \beta_{12}, \tau_{21}^{\text{ref}}, \beta_{21}, \alpha \in [0.1, 0.6]$, with a temperature law $\tau_{ij}(T) = \tau_{ij}^{\text{ref}} + \beta_{ij}(1/T - 1/T_{\text{ref}})$; an optional UNIFAC (UNIQUAC Functional-group Activity Coefficients [5]) group-contribution prior regularises the τ's. This is the default and the most expressive activity model.

Single-parameter (`gamma_inf`). Collapsing the NRTL closure to a symmetric one-parameter form yields the analytic infinite-dilution law $\ln \gamma_2(x_2, T) = \ln \gamma_2^{\infty}(T)\,(1 - x_2)^2$, with $\ln \gamma_2^{\infty}$ affine in $1/T$. This is the minimal activity model and the setting in which the identifiability analysis is cleanest (§4).

Differentiable COSMO-SAC. A σ-profile head emits, for each molecule, a 51-bin screening-charge-density profile $p(\sigma)$ (area per bin, summing to the cavity area A). A parameter-free COSMO-SAC layer (Lin–Sandler / VT-2005 conventions) maps the pair of profiles to $\ln \gamma_2$ by a damped fixed point for the segment activity coefficient,

$$\Gamma(\sigma_m) = \Big[\textstyle\sum_n p_{\text{norm}}(\sigma_n)\, \Gamma(\sigma_n)\, \exp\big(- \Delta W(\sigma_m, \sigma_n)/RT \big) \Big]^{-1}, \tag{6}$$

with a precomputed exchange-energy matrix ΔW (electrostatic misfit plus a hydrogen-bonding term); the restoring contribution is $\ln \gamma_2^{\text{res}} = (A_2/a_{\text{eff}}) \sum_m p_2^{\text{pure}}(\sigma_m)[\ln \Gamma_{\text{mix}}(\sigma_m) - \ln \Gamma_2^{\text{pure}}(\sigma_m)]$, and an optional Staverman–Guggenheim combinatorial term uses the molecular cavity volume and area. The combinatorial size factor $r_i = V_i/r_0$ takes the *true* COSMO cavity volume (V_{cosmo}) so that it shares the cavity basis of the area A_i. This closure — COSMO-SAC evaluated over a network- *predicted* σ-profile, and the intervention that swaps that profile for the measured one — is the object of study in the grounding analysis (§6.1).

5.5 The SLE solver and implicit differentiation

Given $\Phi(T)$ and a closure for $\ln \gamma_2$, the saturated mole fraction solves $\ln x_2 = -\Phi(T) - \ln \gamma_2(x_2, T)$, a fixed point in x_2 because $\ln \gamma_2$ depends on composition. We iterate a damped map

$$x_2^{(k+1)} = \lambda \exp\big(- \Phi(T) - \ln \gamma_2(x_2^{(k)}, T) \big) + (1 - \lambda)\, x_2^{(k)}, \tag{7}$$

with closure-specific kernels for the NRTL, `gamma_inf`, and COSMO-SAC paths. Rather than back-propagating through the iterations, gradients use the implicit-function theorem at the converged $x_2^\star$,

$$\frac{\partial \ln x_2^\star}{\partial \theta} = - \frac{\partial(\Phi + \ln \gamma_2)/\partial \theta}{1 + x_2^\star\, \eta}, \qquad \eta = \frac{\partial \ln \gamma_2}{\partial x_2}, \tag{8}$$

which is exact at the fixed point, memory-constant in the iteration count, and exposes an explicit stability criterion: the gradient is well conditioned unless $1 + x_2^\star \eta \to 0$. Iteration counts (train 16, eval 30) were set by a convergence audit — a shorter n=8 schedule failed a 5.86-$\ln x_2$ convergence test on strongly non-ideal pairs (SI §A).

5.6 Adaptive correction and the grounding intervention

A bounded, gated "adaptive physics correction" perturbs the physical parameters $(T_m, \Delta H_{\text{fus}}, \tau)$ rather than the output, re-solves (7), and blends $\ln x_2^{\text{final}} = \ln x_2^{\text{phys}} + (1-w)\,\text{clip}(\Delta)$; keeping the correction in parameter space preserves the physical decomposition. The *grounding intervention* (a σ-oracle) replaces the predicted σ-profile $\hat{z}$ with the true VT-2005 profile $z^\star$ before the closure, matched by canonical SMILES. It is applied three ways to rule out wiring artifacts: at evaluation only, injected at training time so the rest of the model co-adapts, and as a channel-swap control; an analogous override substitutes true crystal targets. The activity target used for the closure/insufficiency decomposition (§6.2) is evaluated at infinite dilution, $\ln \gamma_2^\infty = \lim_{x_2 \to 0} \ln \gamma_2$, at 298 K.

5.7 Training: three-phase curriculum and single-component grounding

Optimisation follows a fixed three-phase curriculum: (P1) auxiliary-property pretraining on crystal, Hansen, and $\ln \gamma_2^\infty$ targets with no SLE loss; (P2) full SLE training, with the adaptive correction unfrozen partway and an optional crystal→pair weight ramp; (P3) low-learning-rate stabilisation with monotonicity and van't Hoff regularisers. External single-component data enter through *grounding streams*: each stream is a sidecar of *self-solvent* rows (solvent = solute, no solubility label, only the target factor's mask active), interleaved into P1/P2 at a configurable step budget, with a mandatory scaffold-leak guard excluding any pool molecule whose scaffold is in validation or test. The reference streams ground the crystal factor (melting-point pool) and the activity factor (σ-profile database); this single-component-grounding pattern is the mechanism by which an otherwise non-identifiable factor (§4) is pinned by abundant external data a black-box predictor cannot use.

5.8 Objective

The training loss is a weighted sum with per-phase weights. Only a small set of terms is load-bearing for the claims in this paper: a bin-weighted Huber on $\ln x_2$; auxiliary losses on the σ-profile (an earth-mover distance), on crystal $T_m/\Delta H_{\text{fus}}$, on $\ln \gamma_2^\infty$, and on Hansen parameters; and structural regularisers for temperature monotonicity and the van't Hoff slope. The remaining terms (channel orthogonality, contrastive, mixture-of-experts balance, and various priors) are auxiliary; the complete taxonomy and weights are tabulated in the SI (§A).

5.9 The DirectGNN control

DirectGNN shares the encoder, interaction, readout, and pair representation, adds a thermometer temperature encoding, and maps directly to $\ln x_2$ with a bin-weighted Huber loss and no heads, closure, or solver. Because it shares everything but the thermodynamic bottleneck, the TGNNSolv−DirectGNN gap is attributable to the bottleneck rather than to representation, and DirectGNN is the reference for every "physics tax" comparison in §6.

5.10 Reproducibility

All controlled comparisons are run at three seeds and reported as mean±sd on a cross-arm intersection lock (rows supervised and finite in every arm). Solver iteration counts, COSMO-SAC constants, the exact feature list, per-phase loss weights, and the compute ledger (which results are

Figure 5: Controlled comparison on the scaffold split (3 seeds, mean±sd, n=5608; metrics are the 3-seed mean of the cross-arm intersection lock — rows supervised and finite in every arm — from `run_e5_comparison.py`). DirectGNN shares the encoder but has no thermodynamic closure; the physics closures trail it (the "physics tax"). Replacing the model's learned σ-profile with the *true* VT-2005 profile (the σ-oracle, salmon) gives the *worst* arm, collapsing R^2 to ≈ 0 (-0.03 ± 0.04): by MAE the model predicts solubility distinctly better from its invented σ (1.85) than from the measured one (2.25). Per-arm exact values are transcribed in the SI.

full-corpus GPU runs versus CPU-reproducible proxies) are given in the SI (§A, §F); the named analysis scripts regenerate every figure and table from committed artifacts.

6 Results

6.1 The grounding paradox

Feeding true σ-profiles into the closure degrades it rather than improving it, in both the activity and the solubility target, and it fails the way a *misspecified map* does—not the way missing information would. In the solubility target the effect erases any positive R^2 (Fig. 5): the model predicts solubility better *from its own learned σ-profile* than when *given the true measured one* (MAE 1.85 vs 2.25, in $\ln x_2$ units). Two facts pin the cause on the map. First, restoring the physically *more* complete Staverman–Guggenheim combinatorial term (the entropic contribution from solute–solvent size and shape mismatch) *increases* error—what a wrong map does, not what a missing input would. Second, on the activity target ($\ln \gamma_2^\infty$, n=60 matched pairs) the fixed true-input closure sits about 4× above the irreducible-variance floor a flexible regressor reaches (§6.2), under-predicting $\ln \gamma_2^\infty$ with a systematic bias (Fig. 6). The paradox survives three controls—the evaluation-time oracle, injection of the true profile *at training* (MAE 1.98), and a channel-swap (2.08), all worse than the learned-σ arm—ruling out a wiring artifact. The σ-oracle per-arm values are in §C; the true-at-train and channel-swap figures are the inline values above.

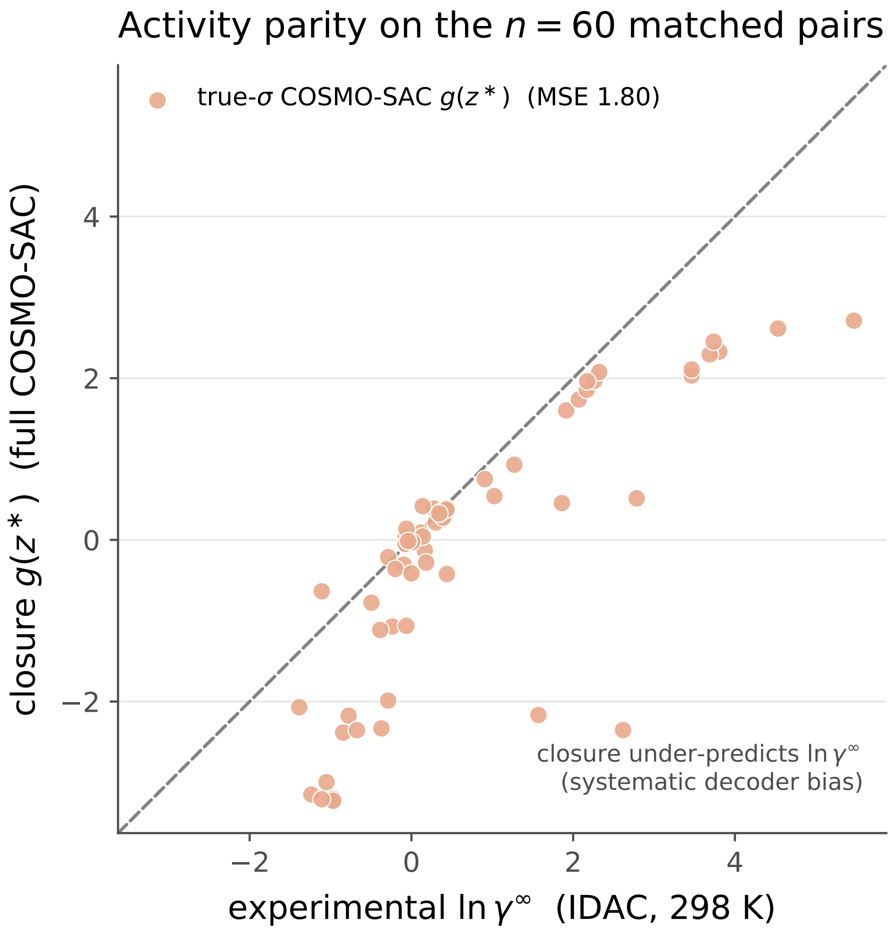


Figure 6: Activity-coefficient parity on the n=60 matched pairs. The true-σ COSMO-SAC closure $g(z^\star)$ (salmon) falls systematically below the diagonal at larger $\ln \gamma_2^\infty$, under-predicting the activity coefficient (MSE 1.80, full convention). This systematic downward bias is the visible form of the decoder misspecification B_{closure}; for reference a flexible regressor of $\ln \gamma_2^\infty$ on the same true profiles reaches the irreducible-variance floor (MSE $\approx$ 0.40, our B_{insuff} estimator; §6.2), so the closure's excess over that floor is B_{closure}.

6.2 Measuring the ceiling: the closure dominates

We apply the decomposition of §4.2 to the n=60 matchable activity pairs (298 K, infinite dilution), with $z^\star$ the true concatenated solute/solvent σ-profiles and m the experimental $\ln \gamma_2^\infty$; Table 2 reports the bounds. The set spans 41 solutes and 17 solvents in 102 profile dimensions, with two formamide pairs dropped on a canonical-SMILES tautomer mismatch.

We grade the evidence in two tiers: *Tier 1* claims hold for *any* COSMO-SAC configuration and are the load-bearing ones, while *Tier 2* claims are configuration-robust corroboration that is not strictly independent of the model fit.

The load-bearing separation (convention-independent, Tier 1). Because $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^\star)]$ depends only on $(m, z^\star)$, its upper bound is the same in every closure convention; subtracting it from the total therefore lower-bounds B_{closure} even for the model's *deployed* residual-only convention, giving $B_{\text{closure}} \gtrsim 0.91$, well above $B_{\text{insuff}} \lesssim 0.62$ (Table 2). This is the claim we lean on, and it

Table 2: Closure–insufficiency decomposition on the true-input activity path. Bounds, not a point split (Lemma 3). Convention "full" = COSMO-SAC with the combinatorial term (size factor $r_i = V_i/r_0$ on the true COSMO cavity volume, VT-2005); "res" = residual-only (the deployed default). B_{insuff} upper bounds are convention-independent estimates of $\mathbb{E}[\text{Var}(m \mid z^\star)]$; "kNN" is a direct high-dimensional estimate (biased upward), the others are law-of-total-variance / out-of-fold upper bounds. The *load-bearing* separation is the convention-independent $B_{\text{closure}} \geq \text{MSE} - B^{\text{up}}_{\text{insuff}}$ row, which exceeds B_{insuff} under *both* conventions ($1.24, 0.91 > 0.62$); the assumption-free constant-offset (Jensen) bound separates only under "full" (0.72) and inverts under the deployed "res" ($0.43 < 0.62$).

Quantity	full	r
$\text{MSE}_{\text{total}} = B_{\text{closure}} + B_{\text{insuff}}$ (exact)	1.80	1.4
B_{closure} lower via $\text{MSE}{-}B^{\text{up}}_{\text{insuff}}$ (RF; B_{insuff} *conv.-indep., load-bearing*)	1.24	0.9
B_{closure} lower bound (Jensen, offset^2; assumption-free; *full-only*)	0.72	0.4
B_{insuff} upper (bin g; law of total variance)	0.53	0.6
B_{insuff} upper (RF / Ridge out-of-fold)	0.56/0.62	sam
B_{insuff} (kNN-in-$z^\star$, convention-indep., biased up)	0.40	0.4
label noise σ^2_η (IDAC, unestimable here)	assumed 0 (no replicates in matched se	

needs no smoothness assumption beyond the validity of the B_{insuff} upper bound. We stress-test the separation directly (App. C): it certifies under the k-NN, binning, and random-forest insufficiency estimators, is stable to injected label noise, and survives leave-one-solvent-out under the principled law-of-total-variance bound (17/17 folds), though it weakens under looser estimators—so we present it as a *supporting*, low-γ-regime measurement rather than a stand-alone certificate. Under the full COSMO-SAC convention the separation *sharpens* to an assumption-free constant-offset bound: a constant decoder offset alone accounts for $B_{\text{closure}} \geq 0.72$ (Fig. 7), exceeding every B_{insuff} upper bound (≤ 0.62), and subtracting the insufficiency bound gives $B_{\text{closure}} \geq 1.24 > 0.62$ ($\sim 2\times$). That constant-offset bound is convention-specific and *inverts* under the deployed residual-only default ($0.43 < 0.62$; hedge 1), which is exactly why we headline the convention-independent bound above rather than the constant offset. Label noise is unestimable in this replicate-free set and assumed zero; the bounds we rely on are invariant to zero-mean noise, so the ceiling is not label noise either.

Convention-robust corroboration (Tier 2). The genuinely convention-robust fact is that adding the physically *more* complete Staverman–Guggenheim combinatorial term *increases* error (activity MSE $1.47 \to 1.80$)—what a misspecified map does, not what missing information would. This is robust to the size-factor's volume basis: recomputing the combinatorial term on the true COSMO cavity volumes (VT-2005 V_{cosmo}) rather than an RDKit molar volume leaves it essentially unchanged (Jensen $0.714 \to 0.717$, MSE $1.793 \to 1.800$), because the term depends on molecular-size *ratios* in which a near-uniform volume rescaling cancels. A second observation is consistent but is *not* independent of the decomposition: a flexible regressor of m on $z^\star$ fits the activity target $\sim 4\times$ better than the fixed true-input closure—but that regressor is exactly our convention-independent B_{insuff} estimator (≈ 0.4), so "$4\times$ better" restates that the closure sits far above the irreducible-variance floor (that B_{closure} is large) rather than corroborating it by an independent route. We therefore lean on the dimension-robust random-forest / ridge out-of-fold and binning bounds and on the combinatorial effect, which agree that B_{closure} exceeds B_{insuff}.

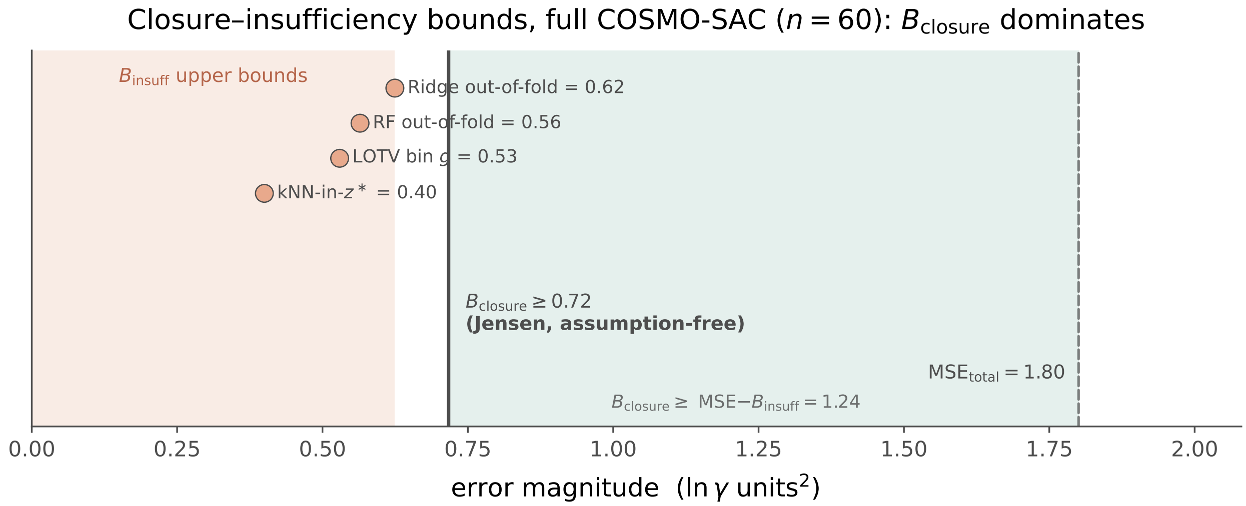


Figure 7: One-sided bounds on the closure–insufficiency split (full COSMO-SAC convention). The assumption-free Jensen lower bound on the decoder misspecification, $B_{\text{closure}} \geq 0.72$, exceeds every valid upper bound on the input insufficiency $B_{\text{insuff}} \leq 0.62$ (four independent estimators cluster in the salmon region); the white gap is a separation that holds with no smoothness assumption. The teal region is the closure share of the total error (MSE = 1.80). This assumption-free separation is convention-specific; under the deployed residual-only default it rests instead on the convention-independent $B_{\text{closure}} \geq \text{MSE} - B^{\text{up}}_{\text{insuff}} \approx 0.9 > 0.6$ bound (§6.2).

Scope and hedges. An adversarial audit sharpened the scope without inverting the verdict. The constant-offset bound is convention-specific—it inverts under the deployed residual-only default, where the convention-independent bound ($B_{\text{closure}} \geq 0.91 > 0.62$) and the combinatorial fact carry the claim instead; the closure error is chiefly a between-solvent bias (a solvent-conditional lower bound $B_{\text{closure}} \geq 0.89$); the solvent-clustered confidence is $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.85$; and the measurement is confined to the low-γ, 298 K, VT-2005-matchable corner, so the numeric ceiling is not claimed to transfer to the finite-composition regime. The four hedges are stated in full in App. C.

6.3 Closure fidelity sets the true-input penalty

To isolate the mechanism behind the grounding paradox we built a controlled, reproducible synthetic experiment in which the only quantity that varies is the fidelity of the closure, with input quality, sample size, and estimator held fixed. A teacher draws a latent $z^\star$ and generates a target through a known map T; a fixed candidate closure g_F reconstructs the target from $z^\star$ under a tunable misspecification that deforms T by a controlled amount, summarized by a scalar fidelity F ($F = 1$ recovers T exactly; F decreases as the closure is pushed further away). The measurement is the true-input penalty $\Delta R^2 = R^2_{\text{oracle}} - R^2_{\text{head}}$, the change in fit from feeding the genuine latent through g_F rather than fitting a free head—B_{closure} made visible, since the free head recovers $\mathbb{E}[m \mid z^\star]$ while g_F is the misspecified map. As F falls the penalty grows monotonically and without exception: averaged over six misspecification functional forms (linear, quadratic, cubic, absolute-value, sinusoidal, cross-term), $F = 1.00$ gives $\Delta R^2 \approx 0$; $F = 0.76$ gives ≈ -0.04; $F = 0.38$ gives ≈ -0.33; and an over-rotation stress test ($F = -0.50$) gives ≈ -2.0 (Fig 8). The ordering is identical across four unrelated teacher families—linear, monotone-nonlinear, kinetics-exponential, and

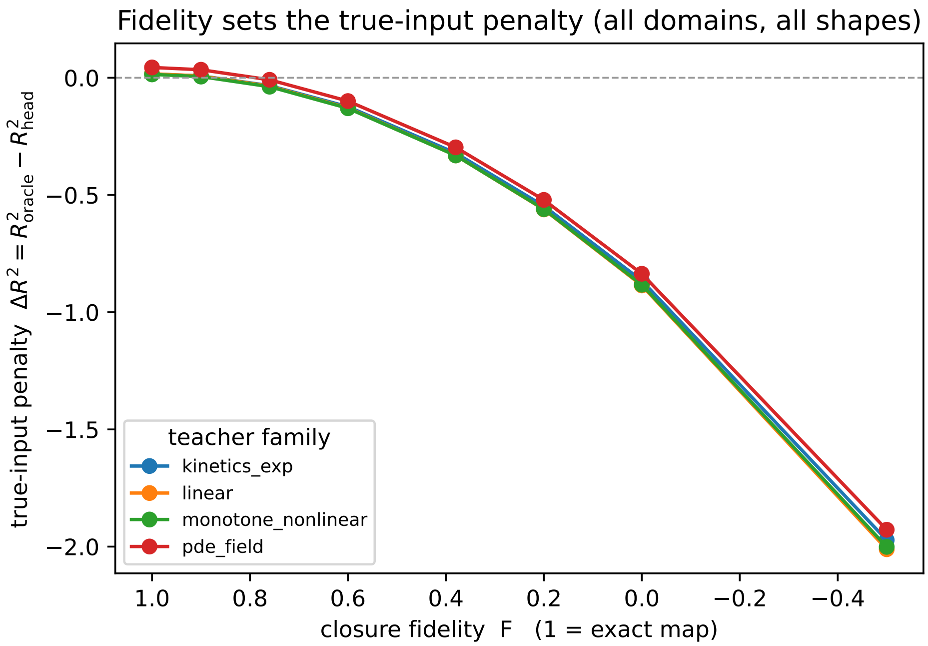


Figure 8: Reproducible closure-fidelity dial (`run_fidelity_dial.py`). As the closure is made more misspecified (fidelity F decreasing from 1), the true-input penalty $\Delta R^2 = R^2_{\text{oracle}} - R^2_{\text{head}}$ grows monotonically, and the same ordering holds across four non-chemistry teacher families (linear, monotone, kinetics-exponential, PDE-field) and six misspecification functional forms. The effect is a property of the fidelity magnitude, not of any one domain or deformation. The real COSMO-SAC closure sits well inside the misspecified regime ($F \lesssim 0.4$)—the sign and rough magnitude of the empirical paradox, not a precise fidelity value.

a PDE-field surrogate, agreeing to within a few hundredths at each F—so the effect is a property of the fidelity magnitude, not of any one domain or deformation form.

This construction separates the two error channels the physics path confounds. When the closure is exact the true input is optimal and $\Delta R^2 = 0$; the penalty appears only as the closure becomes misspecified, and its size is set by that misspecification alone. Placing the real system on the same dial only illustratively—the mapping from a real closure to a scalar F is a synthetic construction, not an independently calibrated measurement—the COSMO-SAC closure over predicted σ-profiles falls well inside the misspecified regime, where the dial's solubility ΔR^2 between the σ-oracle and learned-σ arms (≈ -0.36) lands at $F \lesssim 0.4$ on the monotone curve. This locates the empirical grounding paradox on a mechanistic axis: the loss from feeding true physical inputs is what one expects from a closure of this fidelity, consistent with B_{closure} dominating B_{insuff}.

The same engine also makes concrete why the grounding gap is a symptom rather than a certificate (§4.3). With a knob on the sufficiency of $z^\star$, we can hold the closure *well specified* ($B_{\text{closure}} = 0$) and vary how much target-relevant information the intermediate discards. When $z^\star$ is sufficient (A2 holds) the grounding gap Γ tracks B_{closure} as fidelity varies, the regime in which Theorem 1's proxy is exact; when $z^\star$ is lossy but the closure is correct, $B_{\text{closure}} = 0$ throughout yet $\Gamma > 0$ and grows with the discarded variance (Fig 9). A positive grounding gap can therefore be pure input insufficiency, not closure error—so the certificate of closure-dominance is the convention-independent decomposition, and Γ only corroborates it.

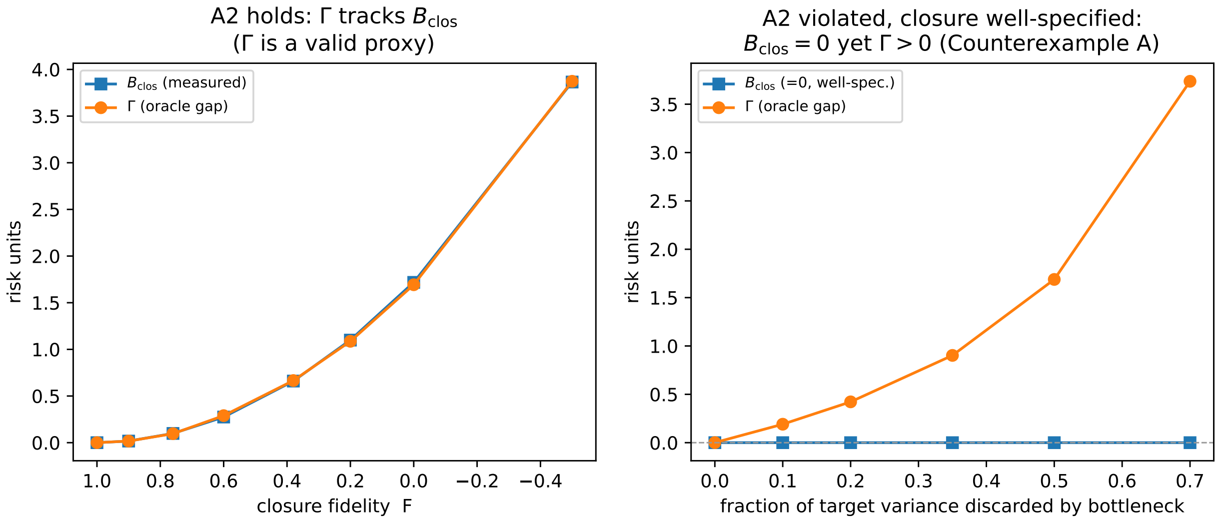


Figure 9: The grounding gap Γ is a symptom, not a certificate (§4.3; `run_fidelity_dial.py`). *Left:* when the intermediate is sufficient (A2 holds), Γ tracks the closure bias B_{closure} as fidelity varies, so Theorem 1's proxy is valid. *Right:* with a *well-specified* closure ($B_{\mathrm{closure}} = 0$) but a lossy intermediate (A2 fails), $\Gamma > 0$ and grows with the discarded variance—a positive grounding gap from pure input insufficiency (the worked Counterexample A of App. B). Hence certification of closure-dominance rests on the decomposition, which needs neither assumption.

6.4 The ceiling is not fixable downstream (Gate B)

If the closure binding were merely a bad point estimate, a learned output-space correction on top of the physics prediction should recover accuracy. It does not. An additive bounded residual with a learned confidence gate ($\hat{y} = \hat{y}_{\mathrm{phys}} + (1 - c)\, r$, with $y = \ln x_2$) leaves the valve effectively shut (applied residual ≈ 0). Forcing the gate open removes the systematic *bias* ($+0.45 \to +0.06$) but does *not* recover *accuracy* (R^2 $0.283 \to 0.274$, essentially unchanged). This is the classical calibration-versus-accuracy separation: when the upstream map is misspecified, a monotone or affine post-hoc map can fix reliability but not resolution. Recalibration buys calibration without buying accuracy, which is what we expect if the closure, rather than the estimate, is the binding constraint.

6.5 A second chemical closure: pKa through a Hammett LFER

The machinery above is not specific to COSMO-SAC. We instantiate it in a second real chemical domain—aqueous pK_a of substituted aromatics—where the fixed closure is a Hammett linear-free-energy relationship (LFER) $g(\sigma) = \mathrm{p}K_{a,0} - \rho\,\sigma$ over the physical intermediate σ—here the Hammett substituent constant, *not* the σ-profile of §3—for which Hansch–Leo tabulated constants provide an external oracle $z^\star$. The analogy is tight: as with the σ-profile through COSMO-SAC, a learnable physical descriptor is pushed through a fixed, approximate, differentiable map. The two classic LFER failure modes fall cleanly onto the two error channels of Lemma 3: *resonance saturation*—a curvature in σ that the linear closure structurally cannot represent—is a mis-map of a function of $z^\star$ and lands in B_{closure}; *ortho/steric* field effects, absent from the tabulated constants, are information the intermediate lacks and land in B_{insuff}. The chemical scope is a fidelity sweep: meta/para monosubstituted benzoic acids, phenols, and anilinium ions are the near-exact ($B_{\mathrm{closure}} \approx 0$) regime, ortho (insufficiency-dominated) and polyfunctional the low-fidelity one.

This decomposition is directly visible before any training. Pushed through the fixed Hammett closure, *p*-nitrophenol reads $\mathrm{p}K_a \approx 8.3$ against an experimental ≈ 7.1 (and *p*-nitroaniline overshoots similarly): the linear closure is off by ~ 1 unit exactly where through-conjugation (σ^-) breaks it—B_{closure} in the wild.

A controlled semi-synthetic construction on real tabulated σ (known ground-truth channels) validates the estimator and reproduces the paper's central distinction in this second closure (Fig. 10). Sweeping closure fidelity with a sufficient intermediate, the grounding gap Γ tracks B_{closure} (the plug-in estimate recovers the truth to within a few percent, e.g. 0.20 vs 0.21, 0.81 vs 0.81), the case where Theorem 1's proxy is valid. Sweeping the ortho/steric channel with a *well-specified* closure gives $B_{\text{closure}} = 0$ throughout yet $\Gamma > 0$ and growing—the pKa realization of the conflation that makes Γ a symptom rather than a certificate. The oracle-σ pipeline (scaffold-resolved substituent$\to$ σ assignment) and this validation extend directly to real data. On the OPERA experimental $\mathrm{p}K_a$ compilation [15] (6,502 compounds), applying the same decomposition to the fixed Hammett closure yields a *measured* fidelity sweep: on the clean meta/para series (n=161) the closure predicts experimental $\mathrm{p}K_a$ to RMSE 0.63 with B_{closure} below the input-insufficiency floor (per-scaffold systematic offsets ≤ 0.25, so the LFER is well specified there), while the ortho/heteroaromatic regime (n=1008) degrades to RMSE 2.4, dominated by input insufficiency ($B_{\text{insuff}} \approx 4.7$—the tabulated σ carries no ortho information, exactly the a-priori channel assignment) with an additional closure misspecification $B_{\text{closure}} \geq 1.1$ on top. This is the near-exact-to-degraded transition the closure is meant to show, now measured rather than asserted. This places the pKa closure near the phase-diagram boundary ($\Delta_\infty \approx 0$) in its clean regime (Fig. 12). What remains is a trained encoder-vs-DirectGNN accuracy comparison on this set, which would place the operating point on the fidelity axis with a *learned* intermediate alongside solubility.

6.6 Identifiability and compensation

Lemma 1 asserts that the crystal/activity split is not recoverable from solubility data alone. Two diagnostics make this concrete. The first is a numerical Fisher audit of the SLE design, computed on synthetic data and independent of any particular activity closure. We evaluate the Fisher information matrix I of the observation model at a reference operating point and read off the conditioning $\mathrm{cond}(I)$ and the Cramér–Rao lower bound on the standard deviation of ΔH_{fus} under three supervision regimes. Table 3 reports the result at a temperature window of $\Delta T = 40$ K. With solubility measurements only, the design is severely ill-conditioned and the CRLB on ΔH_{fus} is roughly 20,000 J/mol, comparable to the quantity being estimated. Adding a direct T_m label drops the conditioning by nearly two orders of magnitude and cuts the bound to about 3,200 J/mol; supplying T_m and ΔH_{fus} together brings it to about 1,700 J/mol. The ill-conditioning is a property of the design rather than of noise level, and it is closed only by external single-component supervision, which is the mechanism the aux-stream pattern supplies.

Widening or narrowing the temperature window does not rescue the solubility-only case. A ΔT sweep moves the solubility-only conditioning from 1.7×10^3 at a wide window to 2.3×10^5 at 40 K and to 1.9×10^{16} at 10 K, so multi-temperature data alone does not resolve the split; the two parameters remain confounded along the ideal-term direction. Restricting the activity slope rather than adding labels helps, but only partially: it cuts $\mathrm{sd}(\Delta H_{\text{fus}})$ to 8,251 J/mol, still well above the label-supervised regimes. A second, model-side diagnostic must be read with more care, and we keep it out of the load-bearing argument. On the supervised test set, with a group-contribution (Joback) crystal reference, the crystal and activity residuals are strongly anti-correlated, $\mathrm{corr}(\delta\Phi, \delta \ln \gamma_2) = -0.962$. This is an *accounting artifact*, not evidence of a learned split: the two residuals sum to the

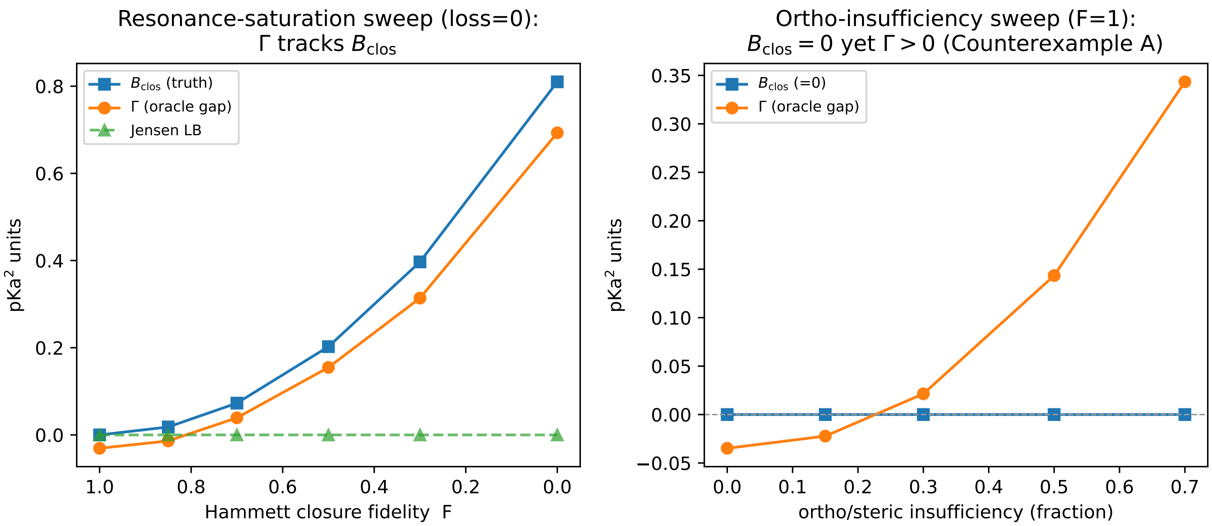


Figure 10: The diagnostic transfers to a second real closure (pKa through a Hammett LFER), on a controlled construction with known ground-truth channels. *Left:* varying closure fidelity with a sufficient intermediate, the grounding gap Γ tracks the closure bias B_{closure} (the assumption-light Jensen bound is flat here because the misspecification carries no mean offset). *Right:* varying an ortho/steric insufficiency channel with a *well-specified* closure, $B_{\mathrm{closure}} = 0$ throughout while $\Gamma > 0$ grows—the pKa realization of the two-cause conflation of §4.3, and the reason certification rests on the decomposition, not on Γ.

Table 3: Fisher audit of the SLE design at $\Delta T = 40\,\mathrm{K}$ (synthetic, closure-independent). Solubility alone leaves the crystal parameter near-unidentified; external single-component labels close the deficiency.

Supervision regime	cond(I)	CRLB sd(ΔH_{fus}) [J/mol]
Solubility only	2.3×10^5	19,895
+ T_m label	5.4×10^3	3,151
+ $T_m + \Delta H_{\mathrm{fus}}$	1.6×10^3	1,689

(negative) $\ln x_2$ fit residual, so their correlation tends to -1 as the fit improves, and a permutation null that destroys any learned coupling still reproduces most of it (-0.90 to -0.93); the model's own factors barely co-vary ($\mathrm{corr}(\hat{\Phi}, \ln \gamma_2) = -0.19$). The large apparent crystal offset ($\delta\Phi \approx -7.1$) is likewise not a usable mis-grounding metric — it is dominated by reference error, since the Joback reference is physically implausible on these drug-like polycyclics (a substantial fraction of predicted T_m exceeds $1000\,\mathrm{K}$), and capping it at a physical bound collapses the offset to ≈ -1.5. We therefore rest the non-identifiability claim on the reference-free Fisher/CRLB audit above, and relegate the compensation figures to the SI (§E) as a cautionary diagnostic rather than a decomposition-quality measurement.

6.7 Grounding the crystal factor: a dose-limited null

A natural test of whether the ceiling is inputs rather than closure is to ground a *different* factor—the crystal term—on abundant external data. We trained the identical model without and with an external single-component crystal-property pool of roughly 15,000 melting points, added through the auxiliary-stream mechanism after excluding any pool solute whose Bemis–Murcko scaffold appears in the validation or test split. The pool did not improve prediction of the crystal properties it supplies (melting-point MAE 45.0 $\to$ 46.6 K, effectively unchanged, if anything marginally worse), and downstream solubility did not improve either ($\ln x_2$ MAE 1.81 $\to$ 1.88; two nominally favorable but slight shifts, in $\ln x_2$ R^2 and the crystal offset $|\delta\Phi|$, do not amount to a gain). We flag this as *dose-limited and inconclusive rather than a clean null.* In this run the crystal auxiliary stream was homeopathically dosed—a small fraction of optimizer steps at low loss weight, so the model saw the external pool less than one full pass at the selected checkpoint—and a stream that did not even lower the MAE of the property it directly supervises cannot license a strong conclusion. What it shows is that *this dilute intervention* does not move the ceiling, not that external crystal labels cannot. We therefore do *not* read it through the activity-path $B_{\mathrm{closure}}/B_{\mathrm{insuff}}$ decomposition: the crystal "closure" is the near-exact ideal-solubility term $\Phi = (\Delta H_{\mathrm{fus}}/R)(1/T - 1/T_m)$, not a misspecified map, so the flat T_m recovery is better read as the crystal/activity split's structural non-identifiability (Lemma 1) and the encoder's transfer limitation (§6.10) than as closure dominance. A properly dosed crystal-grounding test is left to future work.

6.8 Grounding does not pay off in the low-data regime either

The strongest a-priori case for a physics prior is data scarcity: with few training molecules, an inductive bias toward the correct physical factorization should beat a data-hungry black box even when the two tie at full data. We test this directly. Subsampling the training set *by solute* at fractions 0.05–1.0 (the regime that governs scaffold transfer), we train the physics-grounded model and DirectGNN on the identical subsample and evaluate both on the fixed scaffold test (Table 4; seed 42, full-corpus run). The physics arm is *worse at every fraction*, and—against the hypothesis—the gap is *smallest* at the least data (ΔMAE $= +0.08$ at 5%) and *grows* with more ($+1.0$ at full); its R^2 is negative throughout. In the language of Prop. 1, the critical budget $n^\star$ sits below our smallest fraction—the variance a physics prior saves never exceeds its asymptotic bias, even at 5% of solutes, which is a stronger statement than the proposition's bare existence claim. Even in the regime built to favor it, grounding does not buy accuracy; if anything, more data lets the black box pull further ahead. The one metric that inverts is top-1 solvent ranking, higher for the physics arm at 5% (0.42 vs 0.23)—consistent with the ranking-versus-accuracy split of §6.10—but that decision-quality signal does not translate into lower error. This is the last regime in which a physics prior should plausibly have helped; that it does not closes the accuracy case and leaves the structural account of §4 and §6.2—a non-identifiable split fed through a misspecified closure—as the operative explanation.

6.9 Chemistry-resolved structure of the physics tax

Across the full test set the two arms are separated by little more than label noise. On the 3-seed scaffold split the DirectGNN control reaches an $\ln x_2$ MAE of 1.70 ± 0.03, while the grounded physics arm (learned-σ COSMO-SAC closure) reaches 1.85 ± 0.05, for a global $\Delta\mathrm{MAE} = \mathrm{MAE(physics)} - \mathrm{MAE(control)} = +0.14$ (positive meaning the closure adds cost; the per-seed values span $+0.05$ to $+0.25$, so this point estimate is itself only a 3-seed mean, not a tight one; it is computed on unrounded per-seed means, so the rounded 1.70/1.85 above differ by 0.15). At this granularity the physics bottleneck is a small, near-noise tax on accuracy. The interesting structure appears only once the error is resolved along chemical axes, where the average tax turns out to be an average over regimes that disagree in sign.

Table 4: Data-efficiency sweep (train subsampled by solute, fixed scaffold test, seed 42). The physics-grounded arm trails DirectGNN at every fraction; the gap is smallest at the least data and large thereafter. ΔMAE = MAE(physics) − MAE(direct) > 0 means physics is worse. Absolute errors here exceed the headline runs (a lighter per-fraction budget); the valid signal is the within-budget physics−direct gap.

train fraction	MAE		ΔMAE	R^2	
(by solute)	physics	direct	(phys−dir)	physics	direct
0.05	2.28	2.20	+0.08	−0.09	+0.07
0.10	2.44	2.07	+0.37	−0.25	+0.15
0.25	2.93	1.81	+1.12	−0.75	+0.38
0.50	3.12	1.98	+1.14	−0.98	+0.22
1.00	2.98	1.97	+1.01	−0.79	+0.22

Table 5: Per-solvent-class physics tax ΔMAE = MAE(physics) − MAE(control) on the scaffold split (mean ± sd over 3 seeds; positive means the closure adds cost). The intervals are uncorrected per-class pair bootstraps across a family of nine simultaneous comparisons; the two negative (physics-helps) entries are exploratory, not multiplicity-controlled.

Solvent class	ΔMAE (mean ± sd)
Water	$+0.52 \pm 0.35$
Carboxylic acid	$+0.39 \pm 0.16$
Sulfoxide	$+0.26 \pm 0.24$
Nitrile	$+0.21 \pm 0.19$
Alcohol	$+0.15 \pm 0.14$
Aromatic	$+0.03 \pm 0.63$
Hydrocarbon	$+0.02 \pm 0.50$
Halogenated	-0.29 ± 0.36
Amide	-0.22 ± 0.11

The solvent axis is the sharp one. Binning test pairs by solvent class and recomputing ΔMAE per seed (Table 5) shows the cost concentrated in solvents that hydrogen-bond strongly and directionally, precisely the regime in which the COSMO-SAC closure is least faithful and $|\ln \gamma_2|$ is largest. Water carries the heaviest penalty, and the effect decreases monotonically as the donor strength of the class softens. One class inverts the sign: amide solvents, which are aprotic dipolar hydrogen-bond acceptors and therefore closer to the closure's well-specified regime, are the stratum where the tax is smallest and nominally reverses sign (ΔMAE $= -0.22 \pm 0.11$ over seeds). We flag this as exploratory rather than a confirmed win: across a family of nine simultaneous solvent-class comparisons the amide interval is an uncorrected per-class pair bootstrap, and the three-seed spread alone does not exclude zero; the halogenated class shows a nominal improvement of similar magnitude but is not seed-robust. The robust reading is the *direction* — the physics tax shrinks and can turn negative as solvent hydrogen-bond donation weakens toward the acceptor-dominated regime the closure was built for — not the significance of any single class. This solvent-resolved pattern is the chemistry-resolved shadow of the same synthetic dial we used to vary closure fidelity in controlled experiments.

The solute axis tells a compatible but weaker story (Table 6). The closure costs most on

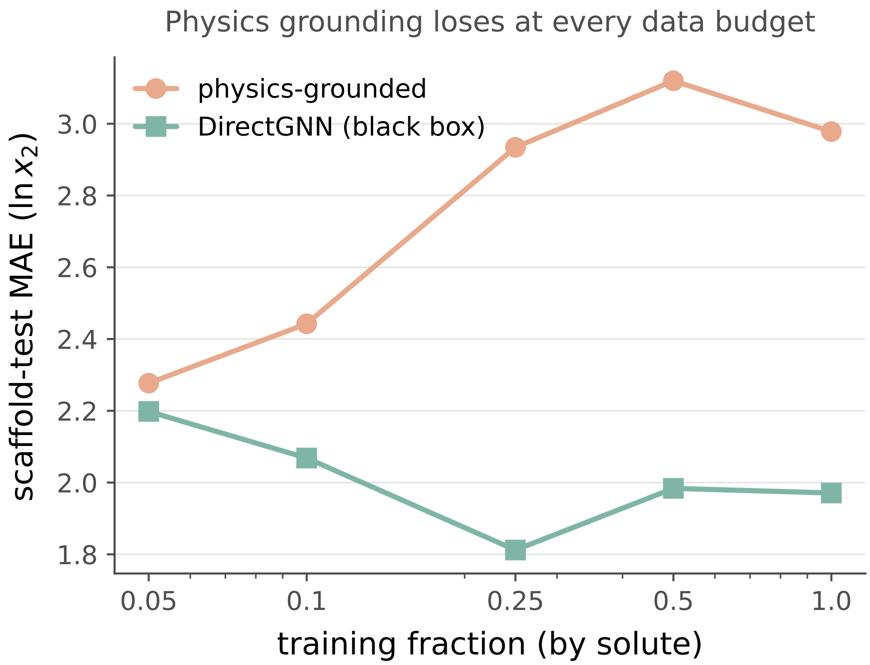


Figure 11: Data-efficiency curve. The physics-grounded arm (salmon) trails the DirectGNN black box (teal) at every training-data fraction; the gap is smallest at the least data and large thereafter—the opposite of the low-data advantage a physics prior is supposed to provide.

ordinary neutral organics—oxygenated solutes, polyaromatics, and heterocycles—while an earlier hypothesis, that it should hurt most on charged, high-$|\ln \gamma_2|$ solutes such as salts, did not replicate once the split was corrected: the charged/salt stratum is neutral. The clean penalty falls on neutral organics, not on the ionic tail we had expected to be hardest.

The most suggestive pattern is a novelty inversion that runs opposite to the usual coverage intuition: the physics arm is neutral on the most novel chemistry (nearest-train Tanimoto ≤ 0.3) and hurts most on the most familiar (> 0.8; Table 6). We flag this as exploratory, not a confirmed effect: the seed spread is not a within-bin sampling interval over which solutes fall in the stratum, and the Tanimoto binning is one of several strata families we examine (solvent class, solute class, similarity) without multiplicity control. Read as a direction rather than a tested quantity, it suggests the ceiling on the physics path is not coverage or extrapolation but the closure's systematic misfit—masked on novel solutes by the large errors both arms already incur there, and exposed on easy, well-covered solutes where the direct control is accurate enough for a structural bias in the closure to become the dominant residual—consistent with the same closure-misspecification reading as the rest of the paper. A multiplicity-controlled, within-bin-bootstrapped test is left to future work. Read together, the maps say the physics decomposition pays where the closure is misspecified relative to the solvent chemistry, and it can only turn into a win in the narrow acceptor-dominated regime the closure was built to describe.

6.10 Supporting diagnostics

The interpretive claims of the preceding sections rest on a set of controlled diagnostics, each run under a named protocol. We report them here with their exact numbers, including the results that run against the physics-informed hypothesis.

Table 6: Per-solute-class and novelty-stratum physics tax $\Delta\mathrm{MAE} = \mathrm{MAE(physics)} - \mathrm{MAE(control)}$ on the scaffold split (positive means the closure adds cost). Solute-class entries are point estimates; the Tanimoto strata carry the 3-seed spread. Exploratory: no multiplicity control across the strata families (§6.9).

Solute stratum	ΔMAE
Oxygenated	$+0.32$
Polyaromatic	$+0.27$
Heterocycle	$+0.17$
Halogenated-aromatic	≈ 0
Charged / salt	-0.04 ± 0.19
Tanimoto ≤ 0.3 (novel)	-0.04 ± 0.18
Tanimoto > 0.8 (familiar)	$+0.51 \pm 0.09$

Aleatoric noise floor. Repeated solubility measurements of the same solute in the same solvent disagree across laboratories, and that disagreement sets a floor below which no model can be expected to fit. Measuring inter-source spread on [11], the mean absolute disagreement is approximately 0.15 $\ln x_2$ units for pairs backed by two sources ($n = 3948$ groups), rising to 0.31 for groups backed by three or more sources ($n = 349$); the corresponding medians are smaller still (0.03 and 0.09). The scaffold-split MAE we report (≈ 1.7 to 1.9) sits well above this band, so label noise does not account for the accuracy ceiling. The floor is small; the ceiling is not a data-quality artifact.

The encoder is transfer-limited, not expressivity-limited. To separate what the frozen graph embedding can represent from what it transfers out of distribution, we fit a linear probe from the encoder representation onto RDKit descriptors and compare in-distribution and held-out fit. The median train R^2 is 0.758 against a test R^2 of 0.448 ($n = 700$ solutes in each partition), a gap of 0.31. Individual descriptors split sharply: FractionCSP3 transfers well (test $R^2 = 0.96$) while HeavyAtomCount does not (0.23). The embedding is expressive enough in sample; it is the generalization of that expressivity across scaffolds that fails. Swapping in a new graph encoder is therefore unlikely to lift out-of-distribution accuracy, and the useful levers are pretraining and coverage rather than architecture.

Temperature extrapolation. A structured van't Hoff activity class (log-solubility affine in $1/T$) encodes the correct temperature dependence of $\ln \gamma_2$ by construction, and this shows up when the split withholds temperature rather than structure. Training on measurements at $T \leq 310$ K and testing at $T \geq 330$ K ($n_{\mathrm{test}} = 3343$), the structured class attains a high-temperature MAE of 0.37 against 1.29 for a random forest over Morgan fingerprints plus temperature (in $\ln x_2$ units), consistent with the temperature-aware $\ln \gamma_2^{\infty}$ modelling of [22]. This is a property of the structured hypothesis class, not a performance claim we make for the trained model; the point is that the functional form carries temperature extrapolation that a descriptor-plus-temperature regressor does not.

Ranking and calibration. Absolute error and decision quality come apart. Despite mediocre MAE, the physics path orders candidate solvents usefully (Table 7; nDCG@3 ≈ 0.73 across $n = 826$ solute groups). Uncertainty, by contrast, is badly miscalibrated in the native model—a nominal-90% interval covers only PICP@90 = 0.13 of held-out points—but split-conformal wrapping repairs

Table 7: Solvent-ranking quality of the physics path on the scaffold test ($n = 826$ solute groups).

Metric	Value
Spearman ρ	0.511
Top-1 accuracy	0.477
Kendall τ	0.436
nDCG@3	0.732

it to 0.92 empirical coverage at the same nominal level. The ranking signal and the post-hoc calibrated intervals are the practically usable outputs, even where the point predictions are not competitive.

7 Discussion

The result is a cautionary measurement for physics-informed modeling: grounding a model on truer physical inputs is only as good as the closure that consumes them, and when that closure is a fixed, mildly misspecified map, better inputs surface its bias rather than help. The external latent does not make the physics path win. What it buys is *attribution*: on the infinite-dilution, low-γ subset we can measure, the ceiling sits in the decoder (B_{closure}), with the inputs (B_{insuff}) ruled out as its source and label noise unestimable but assumed zero (and unable to manufacture the constant offset the bounds rest on). We do not claim this numeric attribution transfers unchanged to the finite-composition, high-γ regime the solubility paradox itself spans; that the solubility paradox shares the same closure-misspecification cause is a conjecture supported by in-regime signs—the more-physics-worse effect on the full n=5608 set and the chemistry map—not a measured fact. For practice this relocates the lever. Within this closure family, effort spent on richer inputs or a better encoder is mis-targeted; the payoff is in the closure itself, for example a corrected or learned residual on the map rather than on its output (which §6.4 shows is inert).

The accuracy case is broad, with an explained boundary. That the physics prior does not improve accuracy is a pattern with no confirmed exception *on accuracy*: the grounded model trails the matched black box on the full corpus (and the external state-of-the-art (SOTA), FastSolv and SolProp retrained on our corpus, does no better—though on a by-solute split that bounds task difficulty rather than ranking head-to-head; App. D), a (dose-limited) attempt to ground the crystal factor on external labels did not move the ceiling (§6.7, inconclusive), and—in the case built to favour a physics prior—it loses at *every* training-data fraction, including the low-data regime where an inductive bias should help most (§6.8). The one axis on which the physical *structure* does help is temperature extrapolation—but as a hypothesis-class property that the trained models do not exploit (§6.10), a partial counterexample we flag rather than hide. Read together these point to a *structural* ceiling: the crystal/activity split is non-identifiable from solubility alone (Lemma 1), and the activity closure is misspecified where we can measure it (§6.2), so within this family—and to the extent our trained instantiations exercise it—better inputs, more data, and auxiliary grounding do not recover an accuracy the factorization cannot express. The contribution is accordingly not an accuracy result but a method—the externally grounded decomposition—for locating such a ceiling, and a proved account of why it sits where it does.

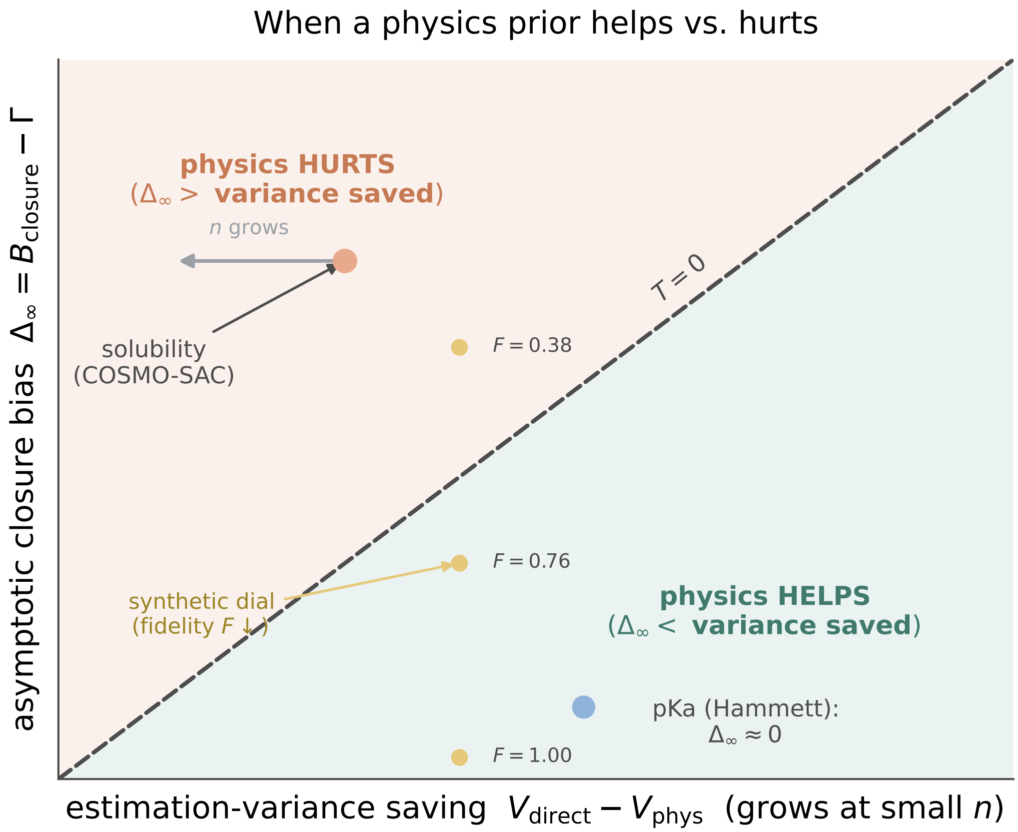


Figure 12: When a physics prior helps versus hurts. The physics tax has sign $T = \Delta_\infty -$ (variance saved): a fixed-closure prior helps only below the $T = 0$ diagonal, where the estimation variance it saves exceeds its asymptotic closure bias $\Delta_\infty = B_{\text{closure}} - \Gamma$. The variance saving shrinks as n grows, so systems drift left: solubility/COSMO-SAC sits deep in the "hurts" region and moves further in with more data, pKa through a Hammett closure sits near the boundary ($\Delta_\infty \approx 0$) in its clean meta/para regime and moves into the tax region for ortho substituents (measured on experimental data, §6.5), and the synthetic dial climbs out of "helps" as closure fidelity F falls. Conceptual—only the sign is measured, not the boundary's exact shape.

When a physics prior must lose. The tax rule (Prop. 1) turns these negatives into a phase picture rather than a single verdict (Fig. 12), and says *when* to expect them. A fixed-closure prior helps only when the estimation variance it saves exceeds its asymptotic bias $\Delta_\infty = B_{\text{closure}} - \Gamma$; both are measurable. The variance benefit is largest at small n and vanishes as data accumulate or as the black box nears the aleatoric floor, whereas Δ_∞ is set by the closure and does not. A prior is therefore predicted to hurt precisely in the mature-data, low-noise, well-covered regime—exactly where our data-efficiency null (a positive tax at every budget, §6.8) and the novelty inversion (the tax heaviest on the best-covered solutes, §6.9) place this system, and where much of modern machine learning operates. The synthetic dial (§6.3) and the pKa closure (§6.5) trace the two governing axes—closure fidelity and intermediate sufficiency—directly. We claim no universal boundary, only a reusable diagnostic: measure B_{closure} through the decomposition, read Δ_∞ off the large-n asymptote, and the sign of the physics tax follows. In this sense the failure is not idiosyncratic to solubility; it recurs whenever a misspecified closure is coupled to an expressive encoder on abundant, low-noise data.

Limitations. (i) B_{insuff} is bounded, never point-identified: there are no true replicate measurements at fixed $z^{\star}$, so the insufficiency estimates lean on a smoothness assumption. The constant-offset (Jensen) bound is smoothness-free but assumes m and g share the $\ln \gamma_2^{\infty}$ (mole-fraction, Lewis–Randall) reference state; we check this two ways—the closure is nearly exact within the largest single-solvent stratum (pyridine, MSE 0.07), which a uniform reference-state offset could not produce, and an independent recomputation reproduces the closure output—so the mean offset is genuine decoder bias, not a units artifact. We headline the convention-independent bound (§6.2) as the load-bearing claim. (ii) The assumption-free constant-offset margin is convention-specific (full COSMO-SAC) and *inverts* under the deployed residual-only default; there the separation rests on the convention-independent bound $B_{\text{closure}} \geq \text{MSE} - B_{\text{insuff}}^{\text{up}}$ and the more-physics-worse effect (which we verified is robust to the molecular-volume basis). (iii) The measurement is at infinite dilution and 298 K on the VT-2005-matchable, low-γ subset; the non-ideal amide-solvent regime is untested. (iv) Absolute scaffold $\ln x_2$ accuracy is encoder-bound and near the aleatoric label-noise floor; this paper is not an accuracy claim.

8 Conclusion

Feeding a physics-informed solubility model its true physical inputs made it worse. What the paper contributes is a way to measure that failure: with an externally grounded latent, the true-input excess risk splits into a decoder-misspecification term and an input-insufficiency term, and on the infinite-dilution, low-γ subset we can measure, the ceiling turns out to be the closure rather than the inputs (label noise unestimable and assumed zero). The practical implication is to fix the map itself, since within this closure truer inputs surface its bias and an output-space correction is inert.

Data and code availability

All code and the intermediate result artifacts needed to regenerate every figure and table are available at `[repository URL]` (commit `[hash]`), released under `[license]`. The solubility corpus is BigSolDB 2.0 [11]; the true σ-profiles are the VT-2005 database, ingested to `results/sigma_profile_artifact/si` the experimental infinite-dilution activity coefficients are compiled from ThermoML (`notebooks/data/raw/idac.c` The Bemis–Murcko scaffold split is produced by `scripts/prepare_data.py`. The controlled comparison (Fig. 5, §6.9) is produced by `scripts/experiments/run_e5_sigma_grounding.sh` and aggregated by `scripts/analysis/run_e5_comparison.py` over `results/e5_sigma_grounding/seed_{42,43,44}/`; the 3-seed aggregate is tabulated in `results/e5_sigma_grounding/THREE_SEED_SUMMARY.md`. The closure–insufficiency decomposition (Table 2, Figs 6 and 7) is regenerated deterministically by `scripts/analysis/run_b_insuff_decomposition.py` → `results/b_insuff/decomposition.json` from the committed matched-pair table `results/b_insuff/matched_pairs.csv` (the n=60 VT-2005-matchable pairs). The second-domain pKa measurement (§6.5) uses the OPERA experimental compilation [15]: `scripts/data/build_pka_hammett_subset.py` assigns the oracle Hammett σ and `scripts/analysis/run_pka_real_decomposition.py` produces the decomposition (→`results/pka_hammett/rea` with the controlled semi-synthetic validation in `scripts/analysis/run_pka_hammett_probe.py`. The synthetic fidelity dial (Fig. 8) is `scripts/analysis/run_fidelity_dial.py`, and the conceptual schematics (Figs 2, 4, 3, 12) and the architecture figure (Fig. 1) are `scripts/analysis/make_concept_figures` The model checkpoints and the full per-arm prediction CSVs are large and are deposited at `[Zenodo DOI]` rather than in git; the named scripts regenerate them from the raw sources. The manuscript compiles with `make` in `paper/`.

Supplementary Information

A Supplementary methods

COSMO-SAC constants. The differentiable COSMO-SAC layer uses the standard VT-2005 / Lin–Sandler constants: effective segment area $a_{\text{eff}} = 7.5\,\text{Å}^2$, electrostatic-misfit constant $\alpha' = 16466.72\,\text{kcal}\,\text{Å}^4\,\text{mol}^{-1}\,\text{e}^{-2}$, hydrogen-bond coefficient $c_{\text{hb}} = 85580$ with cutoff $\sigma_{\text{hb}} = 0.0084\,\text{e}\,\text{Å}^{-2}$, coordination number $z = 10$, and Staverman–Guggenheim normalisers $r_0 = 66.69\,\text{Å}^3$, $q_0 = 79.53\,\text{Å}^2$; the σ-profile grid is 51 bins on $[-0.025, 0.025]\,\text{e}\,\text{Å}^{-2}$. The segment-activity fixed point is damped ($\lambda = 0.7$) and run for 16 iterations at train time and 30 at evaluation.

Crystal head. $T_m = 100+600\,\sigma(\cdot) \in [100, 700]\,\text{K}$, $\Delta H_{\text{fus}} = S_H\,\text{softplus}(\cdot)$; the group-contribution T_m prior is affine- calibrated on the train split before entering the head.

Loss. Only the following terms are load-bearing for the claims: a bin-weighted Huber on $\ln x_2$; auxiliary σ-profile (earth-mover), crystal ($T_m, \Delta H_{\text{fus}}$), $\ln \gamma_2^\infty$, and Hansen losses; and van't Hoff-slope and monotonicity regularisers. The full ∼30-term taxonomy, per-phase weights, graph featurisation, and encoder hyperparameters are in the training configs (`configs/physics_grounded.yaml` and `config.py`); every reported run pins its resolved config in `results/*/resolved_config.json`.

B Proofs

Proof of Lemma 1 (non-identifiability). Write $u = 1/T$ and $h = \Delta H_{\text{fus}}/R$, $u_m = 1/T_m$. At infinite dilution and with $\Delta C_p = 0$ the observable is

$$\mu(u) = -\Phi(u) - \ln \gamma_2^\infty(u) = -h(u - u_m) - (c + d\,u) = -(h + d)\,u + (hu_m - c),$$

using $\ln \gamma_2^\infty(u) = c + d\,u$ (affine in u by hypothesis). Thus μ depends on the parameters $\theta = (\Delta H_{\text{fus}}, T_m, c, d)$ only through the pair $(S, K) = (-(h+d),\ hu_m - c) \in \mathbb{R}^2$. The activity block spans this plane, $\partial(S, K)/\partial c = (0, -1)$ and $\partial(S, K)/\partial d = (-1, 0)$; hence for *any* crystal value (h, u_m) and any target (S, K) there is an activity pair $d = -S - h$, $c = hu_m - K$ reproducing it. The likelihood level set therefore contains the full two-dimensional family $\{(h, u_m)\}$, so $(\Delta H_{\text{fus}}, T_m)$ are unidentified. Equivalently, the crystal score $\partial(S, K)/\partial h = (-1, u_m)$ lies in the span of the activity columns, so profiling out (c, d) annihilates it and $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = 0$. At finite dilution $\ln \gamma_2(u) = \ln \gamma_2^\infty(u)\,(1 - x_2)^2$ adds $\ln \gamma_2^\infty(u)\,(2x_2 - x_2^2) = O(x_2)$, which is not affine in u because $x_2 = x_2(u)$; the crystal score then has a component outside the activity span of order x_2, so $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = O(x_2^2) > 0$. □

Proof of Lemma 2 (class-dependent information). This is the semiparametric efficient-score construction [24] (here finite-dimensional and Gaussian, so equivalently the partitioned-information / Frisch–Waugh–Lovell identity). With Gaussian noise of variance σ^2, parameter-of-interest score $v = \partial\mu/\partial\Delta H_{\text{fus}}$, and nuisance tangent space $\mathcal{T}_{\mathcal{H}}$ (the L^2-closure of the span of nuisance scores), the efficient score is the residual $\tilde{v} = v - \Pi_{\mathcal{H}} v$ and the efficient information is $\mathcal{I}_{\text{eff}} = \sigma^{-2}\|v - \Pi_{\mathcal{H}} v\|^2$. Here $v = \partial\mu/\partial\Delta H_{\text{fus}} = -(u - u_m)/R \in \text{span}\{1, u\}$. If $\mathcal{T}_{\mathcal{H}} \supseteq \text{span}\{1, u\}$ then $\Pi_{\mathcal{H}} v = v$ and $\mathcal{I}_{\text{eff}} = 0$, recovering Lemma 1; if $\mathcal{T}_{\mathcal{H}}$ is a proper subspace not containing v (e.g. a fixed slope, $\mathcal{T}_{\mathcal{H}} = \text{span}\{1\}$) then $\Pi_{\mathcal{H}} v \neq v$ and $\mathcal{I}_{\text{eff}} > 0$, with a finite Cramér–Rao bound. □

Proof of Lemma 3 (decomposition and bounds). (a) Identity. Split the residual as $m - g(z^\star) = [m - \mathbb{E}(m \mid z^\star)] + [\mathbb{E}(m \mid z^\star) - g(z^\star)]$, square, and take expectations. The squared terms give

$\mathbb{E}[\mathrm{Var}(m \mid z^\star)] = B_{\text{insuff}}$ and $\mathbb{E}[(\mathbb{E}(m \mid z^\star) - g(z^\star))^2] = B_{\text{closure}}$. The cross term vanishes: since $\mathbb{E}(m \mid z^\star) - g(z^\star)$ is $\sigma(z^\star)$-measurable, the tower rule gives $\mathbb{E}\big[(m - \mathbb{E}(m \mid z^\star))(\mathbb{E}(m \mid z^\star) - g(z^\star))\big] = \mathbb{E}\big[(\mathbb{E}(m \mid z^\star) - g(z^\star))\,\mathbb{E}(m - \mathbb{E}(m \mid z^\star) \mid z^\star)\big] = 0$. *(b) Jensen lower bound.* With $\Delta(z^\star) = \mathbb{E}(m \mid z^\star) - g(z^\star)$, $B_{\text{closure}} = \mathbb{E}[\Delta^2] \geq (\mathbb{E}[\Delta])^2 = (\mathbb{E}[m] - \mathbb{E}[g])^2$ by Jensen, using $\mathbb{E}[\mathbb{E}(m \mid z^\star)] = \mathbb{E}[m]$; it requires only that m and g share a reference state, so that $\mathbb{E}[m] - \mathbb{E}[g]$ is a genuine mean discrepancy. *(c) Law-of-total-variance upper bound.* For any binning $B = \mathrm{bin}(g(z^\star))$, $\sigma(B) \subseteq \sigma(z^\star)$, so conditioning on the coarser σ-algebra cannot reduce expected conditional variance: $\mathbb{E}[\mathrm{Var}(m \mid B)] \geq \mathbb{E}[\mathrm{Var}(m \mid z^\star)] = B_{\text{insuff}}$. *(d) Convention-independence.* $B_{\text{insuff}} = \mathbb{E}[\mathrm{Var}(m \mid z^\star)]$ is a functional of the law of $(m, z^\star)$ alone and does not involve g; hence it is identical across closure conventions, and any $\mathrm{MSE}(g_1) - \mathrm{MSE}(g_2)$ is entirely $B_{\text{closure}}(g_1) - B_{\text{closure}}(g_2)$. □

Proof of Theorem 1 (grounding-gap sandwich). Throughout assume $m, g(z^\star) \in L^2$ and g, every $h \in \mathcal{E}$ measurable. *(a)* $\Gamma \geq 0$. By A1, $\phi^\star \in \mathcal{E}$ is a feasible encoder with $g(\phi^\star(x)) = g(z^\star)$, so $R_{\text{e2e}} = \inf_h \mathbb{E}[(m - g(h(x)))^2] \leq \mathbb{E}[(m - g(z^\star))^2] = R_{\text{orc}}$; the infimum need not be attained. Hence $\Gamma = R_{\text{orc}} - R_{\text{e2e}} \geq 0$. *(b)* $R_{\text{e2e}} \geq R_{\text{free}}$. Every $g(h(x))$ is $\sigma(x)$-measurable, and $R_{\text{free}} = \mathbb{E}[(m - \mathbb{E}(m \mid x))^2]$ is the infimum of $\mathbb{E}[(m-p)^2]$ over *all* $\sigma(x)$-measurable p; an infimum over the subclass $\{g(h(\cdot)) : h \in \mathcal{E}\}$ is no smaller. *(c)* $R_{\text{free}} = B_{\text{insuff}}$ *under A2.* From $\mathrm{Var}(m) = \mathbb{E}[\mathrm{Var}(m \mid x)] + \mathrm{Var}(\mathbb{E}(m \mid x)) = \mathbb{E}[\mathrm{Var}(m \mid z^\star)] + \mathrm{Var}(\mathbb{E}(m \mid z^\star))$ and A2 ($\mathbb{E}(m \mid x) = \mathbb{E}(m \mid z^\star)$, so the two $\mathrm{Var}(\mathbb{E}(\cdot))$ terms agree), the two expected conditional variances agree: $R_{\text{free}} = \mathbb{E}[\mathrm{Var}(m \mid x)] = B_{\text{insuff}}$. *(d) Sandwich and identity.* By Lemma 3, $R_{\text{orc}} = B_{\text{insuff}} + B_{\text{closure}}$. With (c), $\Gamma = R_{\text{orc}} - R_{\text{e2e}} = B_{\text{closure}} - (R_{\text{e2e}} - R_{\text{free}})$, and (b) gives $R_{\text{e2e}} - R_{\text{free}} \geq 0$, so $0 \leq \Gamma \leq B_{\text{closure}}$; equality $\Gamma = B_{\text{closure}}$ holds iff $R_{\text{e2e}} = R_{\text{free}}$. Finally $B_{\text{closure}} = 0$ forces $0 \leq \Gamma \leq 0$, i.e. $\Gamma = 0$; contrapositively, under A1–A2, $\Gamma > 0 \Rightarrow B_{\text{closure}} > 0$. *(e) A2 is necessary for the certificate.* If A2 fails, take $x \in \{a, b\}$ equiprobable, $z^\star \equiv 0$ (so A1 holds), $m(a) = +1, m(b) = -1$, and g with $g(0) = 0 = \mathbb{E}(m \mid z^\star)$ so $B_{\text{closure}} = 0$; an encoder $h^\dagger(a) = 1, h^\dagger(b) = -1$ with $g(\pm 1) = \pm 1$ gives $R_{\text{e2e}} = 0$ while $R_{\text{orc}} = 1$, hence $\Gamma = 1 > 0 = B_{\text{closure}}$. So $\Gamma > 0$ alone does not certify $B_{\text{closure}} > 0$; the decomposition (Lemma 3) does. □

Proof of Proposition 1 (sign of the physics tax). Write $D(n) := V_{\text{direct}}(n) - V_{\text{phys}}(n)$, so $T(n) = \Delta_\infty - D(n)$ with $\Delta_\infty = R_{\text{e2e}} - R_{\text{direct},\infty}$. *(i)* $T(n) < 0 \iff D(n) > \Delta_\infty$, immediately. *(ii)* If $\Delta_\infty > 0$ and $D(n) \to 0$, choose $n^\star$ with $|D(n)| < \Delta_\infty$ for all $n > n^\star$ (possible since $D(n) \to 0$); then $T(n) = \Delta_\infty - D(n) > 0$. Monotonicity of D is *not* required for existence of $n^\star$; it is needed only for the stronger claim "$T(n) > 0$ at every n," which we treat as an empirical observation. *(iii)* Since $V_{\text{phys}}(n) \geq 0$, $D(n) \leq V_{\text{direct}}(n)$: the variance a prior can save is bounded by the control's own estimation variance. As the black box approaches the aleatoric floor $V_{\text{direct}}(n) \to 0$, so $\limsup_n D(n) \leq 0$; with both estimators consistent ($V_{\text{phys}}, V_{\text{direct}} \to 0$) one has $D(n) \to 0$ and $T(n) \to \Delta_\infty \geq 0$. When the control is asymptotically Bayes ($R_{\text{direct},\infty} = R_{\text{free}}$) and A2 holds, $\Delta_\infty = R_{\text{e2e}} - R_{\text{free}} = B_{\text{closure}} - \Gamma$ by (3). □

C Per-arm tables and decomposition robustness

Table 8 gives the exact per-arm metrics behind Fig. 5: three seeds (mean±sd) on the cross-arm intersection lock (n=5608, rows supervised and finite in every arm). Both closure families—NRTL and COSMO-SAC over a learned σ-profile—trail the DirectGNN control (the physics tax); the σ-oracle (true VT-2005 profile) is the worst arm, and restoring the Staverman–Guggenheim combinatorial term makes it slightly worse still—the signature of a misspecified map (§6.2).

Table 8: Per-arm scaffold-split metrics (3 seeds, n=5608 cross-arm lock). Ordered by MAE; the physics closures trail DirectGNN and the true-σ oracle is worst. Regenerated by `run_e5_comparison.py` over `results/e5_sigma_grounding/seed_{42,43,44}/`.

Arm	MAE	R^2
DirectGNN (no closure)	1.70 ± 0.03	$+0.42 \pm 0.03$
NRTL closure	1.80 ± 0.07	$+0.34 \pm 0.03$
COSMO-SAC, learned σ (grounded)	1.85 ± 0.05	$+0.33 \pm 0.05$
+ Staverman–Guggenheim combinatorial	1.88 ± 0.09	$+0.32 \pm 0.05$
COSMO-SAC, ungrounded σ	2.04 ± 0.04	$+0.19 \pm 0.01$
COSMO-SAC, true σ (σ-oracle)	2.25 ± 0.02	-0.03 ± 0.04

Robustness of the closure/insufficiency separation. We stress the deployed-convention verdict $B_{\text{closure}} > B_{\text{insuff}}$ (§6.2) three ways on the n=60 set (`run_b_insuff_robustness.py`). *(i) Estimator sensitivity.* The B_{insuff} upper bounds are 0.40 (k-NN), 0.56 (binning/LOTV), 0.62 (random forest), and 0.88 (ridge); since the true B_{insuff} lies below the smallest valid upper bound and $\text{MSE}_{\text{res}} = 1.47$, the verdict $B_{\text{closure}} \geq \text{MSE} - B^{\text{up}}_{\text{insuff}} > B_{\text{insuff}}$ certifies under the three tightest estimators (the ridge bound is loose—non-certifying, not refuting). *(ii) Leave-one-solvent-out.* Dropping each of the 17 solvents in turn, the verdict holds in *all* 17 folds under the law-of-total-variance bound; under the random-forest estimator it holds in only 2/17, because a random forest over-fits conditional variance in 102 dimensions at n=60 (its out-of-fold MSE approaches $\text{Var}(m)$), so its leave-one-out estimate is unstable—an estimator artifact, not a sign flip. *(iii) Label noise.* Injecting zero-mean Gaussian noise $\sigma_\eta \in \{0.2, \ldots, 0.8\}$ into m leaves the estimated B_{closure} lower bound stable ($0.84 \to 0.70$), as expected since $B_{\text{closure}} = \text{MSE} - \mathbb{E}[\text{Var}(m \mid z^\star)]$ is invariant to zero-mean noise. The separation is thus robust under a principled insufficiency estimator and noise-stable, but its numerical strength is estimator-dependent—consistent with the solvent-clustered $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.85$ (hedge 3).

The closure/insufficiency decomposition (full/res conventions; law-of-total-variance (LOTV), random-forest (RF), ridge, and kNN estimators; solvent-clustered bootstrap; the pyridine stratum) is tabulated in §6.2 and regenerated by `run_b_insuff_decomposition.py`.

The four hedges (full). The adversarial audit could not invert the verdict but sharpened its scope:

1. *Convention.* The deployed default is residual-only. Under it the *constant-offset* (Jensen) lower bound inverts ($B_{\text{closure}} \geq 0.43 < B_{\text{insuff}} \leq 0.62$), so that assumption-free bound does not separate, and the $\sim 2\times$ full-convention margin (the sharpened $\text{MSE} - B_{\text{insuff}}$ bound, 1.24 vs 0.62) is a full-COSMO-SAC-only result. What separates under the deployed convention is the convention-independent bound $B_{\text{closure}} \geq \text{MSE}_{\text{res}} - B^{\text{up}}_{\text{insuff}} \approx 0.91 > 0.62$, on which we headline, together with the combinatorial Tier 2 fact.

2. *Confound.* The closure error is predominantly a *between-solvent* systematic bias (~64% of total error); within the largest single-solvent stratum (pyridine, n=20) the closure is nearly correct (stratum MSE 0.07). Because solvent σ is part of $z^\star$, these offsets are still decoder error and give an independent solvent-conditional lower bound $B_{\text{closure}} \geq 0.89$ (full)/0.56 (res).

3. *Confidence.* Honest confidence on the full-convention constant-offset separation is the solvent-

clustered bootstrap $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.85$ (0.83 vs the random-forest upper bound, 0.89 vs the law-of-total-variance bound); we report the solvent-clustered scheme rather than the less conservative i.i.d.-pair one because solvent dependence dominates. The convention-independent margin (0.91 vs 0.62) is a point estimate. The decomposition is checkpoint/seed-independent (the closure has zero learnable parameters).

4. *Selection/scope.* The 60 matchable pairs are the low-to-moderate γ corner (matched mean $\ln \gamma_2^\infty = 0.75$ vs 3.56 for VT-2005-unmatchable pairs, $t = -7.44$; the strongly non-ideal amide-solvent regime is untested because VT-2005 lacks those profiles). This narrows scope but is *conservative*, since the untested regime is a harder closure test. The target is $\ln \gamma_2^\infty$ at 298 K, so the numeric ceiling is not claimed to transfer unchanged to the finite-composition solubility regime.

D External baselines and ablations

We retrained the direct SOTA (FastSolv [1]) and the physics-informed SOTA (SolProp [25]) on our corpus and evaluated them on a by-solute test split (`test_solute.csv`, $n \approx$ 10–12k). In $\ln x_2$ they reach MAE 1.94 (FastSolv, R^2 0.23) and 2.34 (SolProp, calibrated, R^2 −0.18); in $\log_{10} S$, 0.87 and 1.04. Neither beats our matched arms, and the physics-informed SolProp trails the direct FastSolv—the same ordering we find internally between the physics closures and DirectGNN. These runs use a different (by-solute) test file than the scaffold split of the headline comparison, so they bound the task's difficulty rather than rank directly against the scaffold numbers of Table 8; a like-for-like scaffold evaluation of the external models is left to the deposited artifacts. The regime-level reading is that on this corpus no black box, ours or external, escapes the accuracy ceiling—consistent with the structural account of §7. Per-arm ablations (NRTL vs COSMO-SAC, combinatorial on/off) are in Table 8.

E Supplementary mechanism results

Non-identifiability, empirically (the corrupted twin). Training a twin model on data whose crystal labels are corrupted (every T_m shifted by +273 K) leaves the *solubility* fit essentially unchanged—$\ln x_2$ MAE 1.835 versus 1.812 on the corrected data—while the recovered crystal grounding differs by hundreds of kelvin (T_m MAE 258 K versus 45 K). Two very different crystal/activity splits produce the same solubility, which is Lemma 1 made concrete.

Crystal grounding on physical labels. When the crystal head is supervised on measured melting points it predicts T_m to 45 K MAE, about 4× better than a Joback group-contribution reference—so the flat downstream result of §6.7 is a dose/identifiability effect, not an inability to learn crystal properties.

The $\Delta C_p = 0$ approximation. The ideal term omits the constant-heat- capacity correction; a group-contribution audit finds a nonzero ΔC_p for 1441 of 1525 solutes (108,287 rows), so this is a bounded model-form approximation. It cannot produce either headline result. The closure/insufficiency decomposition is measured on the activity target ($\ln \gamma_2^\infty$ at 298 K), where ΔC_p does not enter at all; and the grounding paradox is an arm-to-arm difference (learned-σ vs. oracle-σ) whose two arms share the *identical* crystal term, so any ΔC_p error cancels exactly in the comparison. It is thus orthogonal to the activity closure, not a source of the paradox.

Fusion-supervision scarcity. Direct ΔH_{fus} labels are extremely sparse: 1279 training rows across 31 solutes, and essentially none in validation or test. This is the quantitative backbone of the identifiability argument—there is almost no signal with which to pin the crystal term from data.

NRTL-branch collapse. In the trained NRTL closure the interaction parameter is numerically dead (std(τ_{12}) $\approx 4.8 \times 10^{-5}$): the activity branch degenerates rather than learning a temperature-resolved interaction, a companion to the COSMO-SAC misspecification on the other closure.

Structured temperature extrapolation. Withholding *temperature* rather than structure (train at $T \leq 309.75\,\text{K}$, test at higher T; $n = 3343$), the affine-in-$1/T$ hypothesis classes extrapolate far better than an unstructured descriptor regressor (Table 9); the van't Hoff class also predicts the *direction* of the high-T shift correctly 99% of the time versus 40% for the random forest. This is a property of the structured hypothesis class—the trained neural models are weaker than the van't Hoff baseline—not an accuracy claim for our model.

Table 9: Same-pair high-temperature extrapolation ($n = 3343$, train $T \leq 309.75\,\text{K}$). Structured $1/T$ classes extrapolate; a descriptor regressor and constant baselines do not. "dir." is the fraction of pairs whose high-T shift direction is predicted correctly.

Model	MAE	R^2	dir.
per-pair van't Hoff ($1/T$)	0.37	0.89	0.99
per-pair linear in T	0.41	0.85	0.99
per-pair last-low-T hold	1.20	0.69	0.01
RF (Morgan $+T$)	1.29	0.66	0.40
per-pair mean	1.46	0.56	0.01

F Data provenance and reproducibility

Sources. Solubility: BigSolDB 2.0 [11]. Crystal $T_m/\Delta H_{\text{fus}}$: an open melting-point pool plus a group-contribution prior. True σ-profiles: the VT-2005 database (ingested to `sigma_profiles.csv`, 1432 compounds, with the COSMO cavity volume V_{cosmo} for the combinatorial term). These are quantum-chemically computed for a single reference conformer and protonation state; any mismatch to the experimental measurement conditions is noise in $z^\star$ that inflates B_{insuff} (an imperfect intermediate) rather than B_{closure}, so it is *conservative* for the closure-dominance verdict. Infinite-dilution activity coefficients: ThermoML. Second-domain pKa: the OPERA experimental compilation [15] (`pKa_QR.sdf`, 6502 compounds, public-domain US EPA; cited, not redistributed). The Bemis–Murcko scaffold split is produced by `prepare_data.py`; melting-point tables are auto-detected as °C or K to avoid a +273 K double-conversion.

Compute ledger. The headline σ-grounding comparison (Fig. 5, Table 8) and the data-efficiency sweep (Table 4) are full-corpus GPU runs (three seeds for the former, seed 42 for the latter); the closure/insufficiency decomposition (§6.2), the Fisher audit (§6.6), the noise floor, the encoder probe, and the temperature and mechanism diagnostics of §E are CPU-reproducible from committed artifacts. Each figure and table names the script that regenerates it from `results/`; large prediction CSVs and model checkpoints are deposited separately ([Zenodo DOI]) rather than in the repository.

Seeds and determinism. Controlled comparisons use seeds $\{42, 43, 44\}$ and are reported on a cross-arm intersection lock (rows supervised and finite in every arm); the decomposition is seed-independent (its closure has no learnable parameters).

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